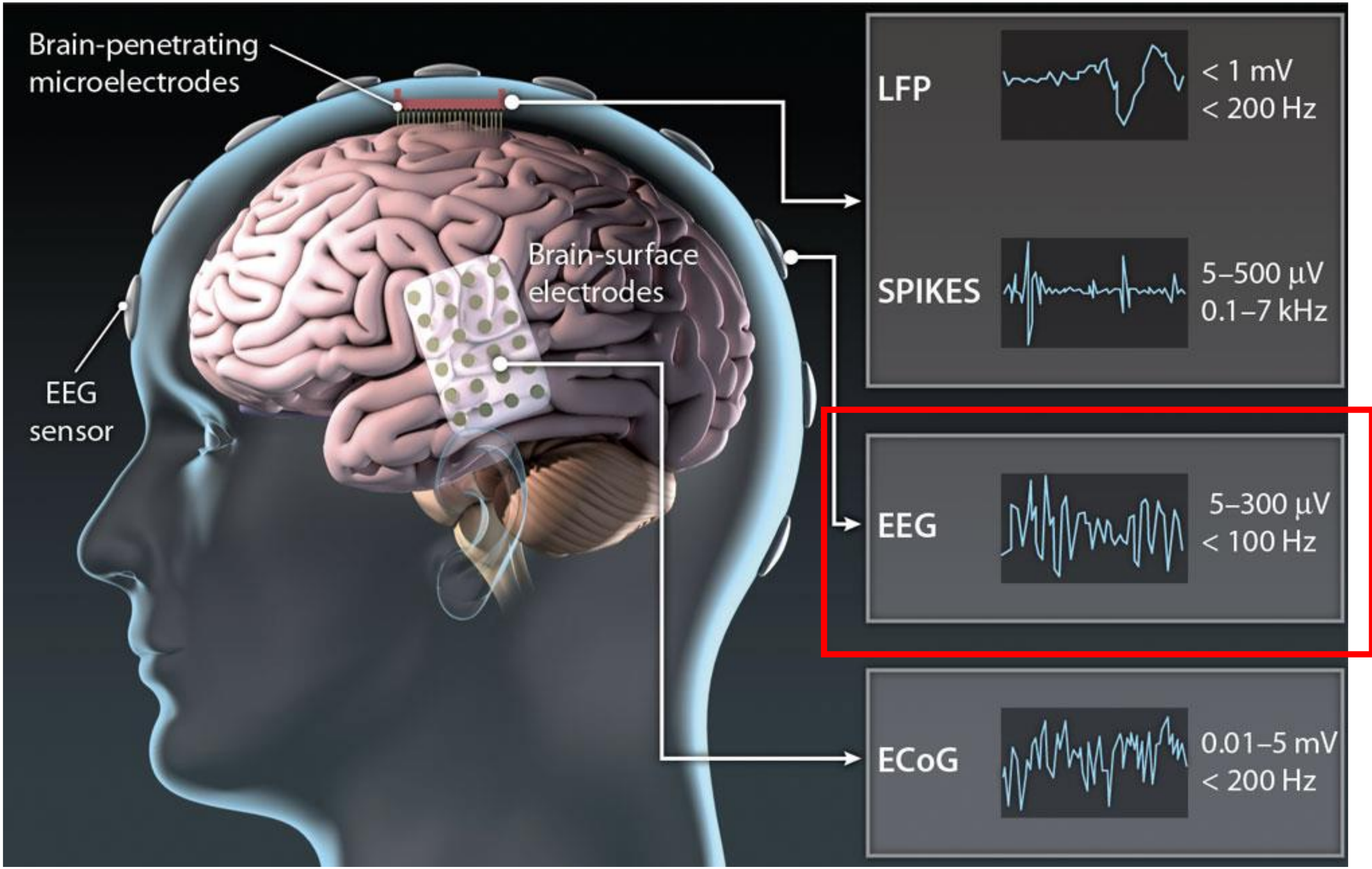
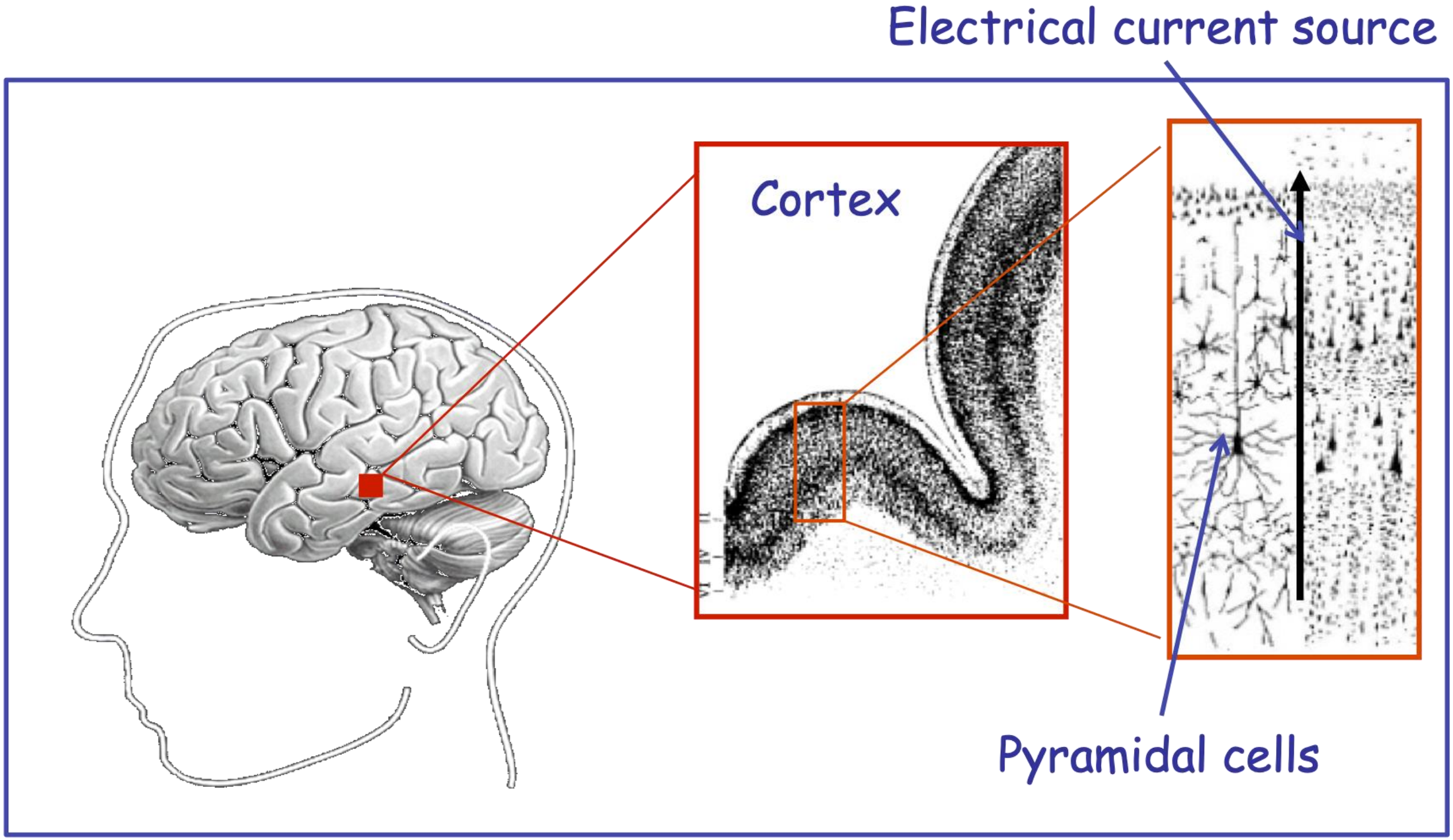
Characteristics of EEG

Electrical Brain Signals



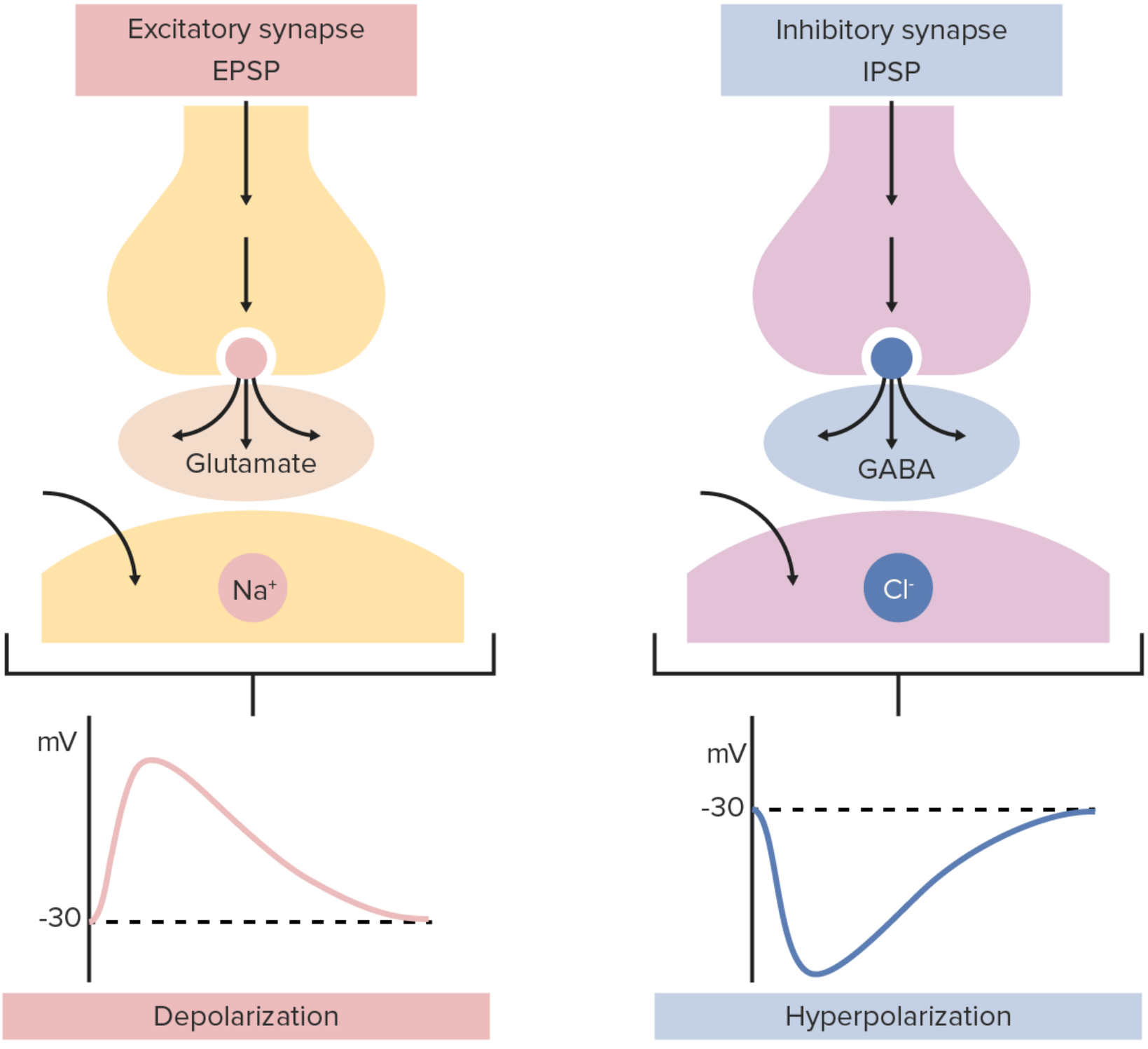
(Thakor, 2013)

Electroencephalography (EEG)



Source: Will Penny

EEG Source


Excitatory synapse
EPSP
Glutamate
Na+
mV
-30
Depolarization
Inhibitory synapse
IPSP
GABA
Cl-
mV
-30
Hyperpolarization

EEG Source

Excitatory PostSynaptic Potential

↓

Ion channels open

↓

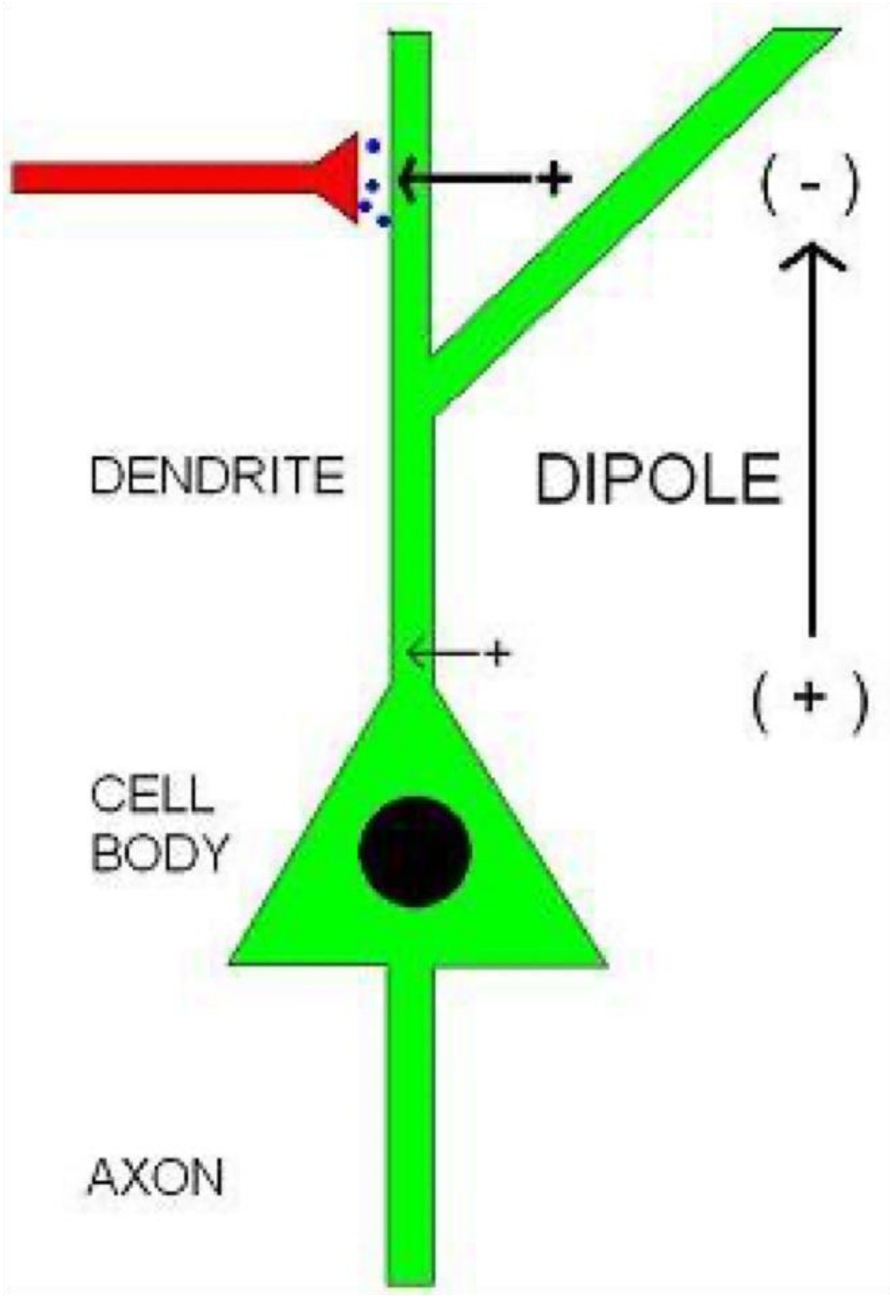
+ Ion channels flow into the cell

↓

More negative outside

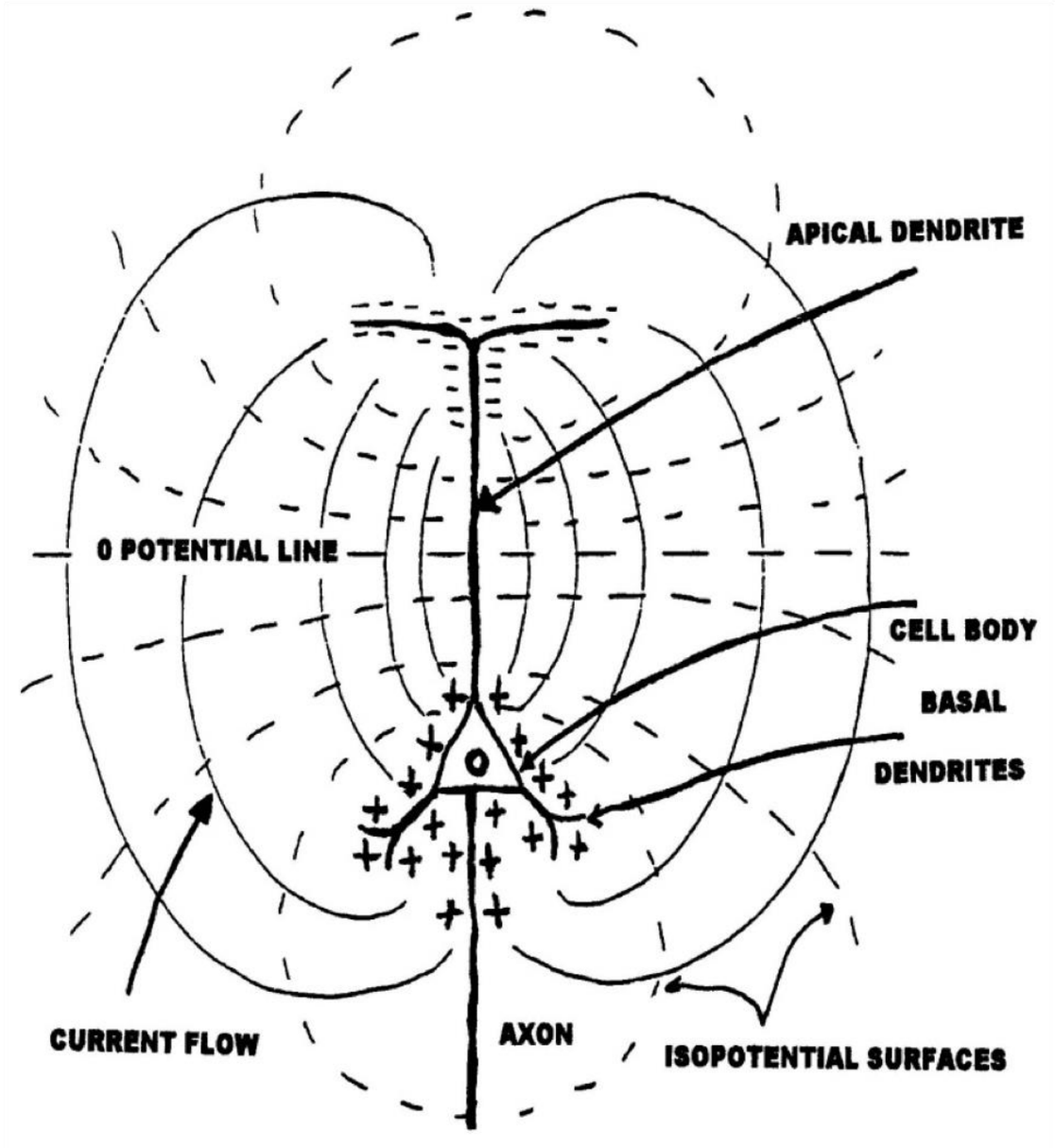
↓

Current flows from soma to dendrites



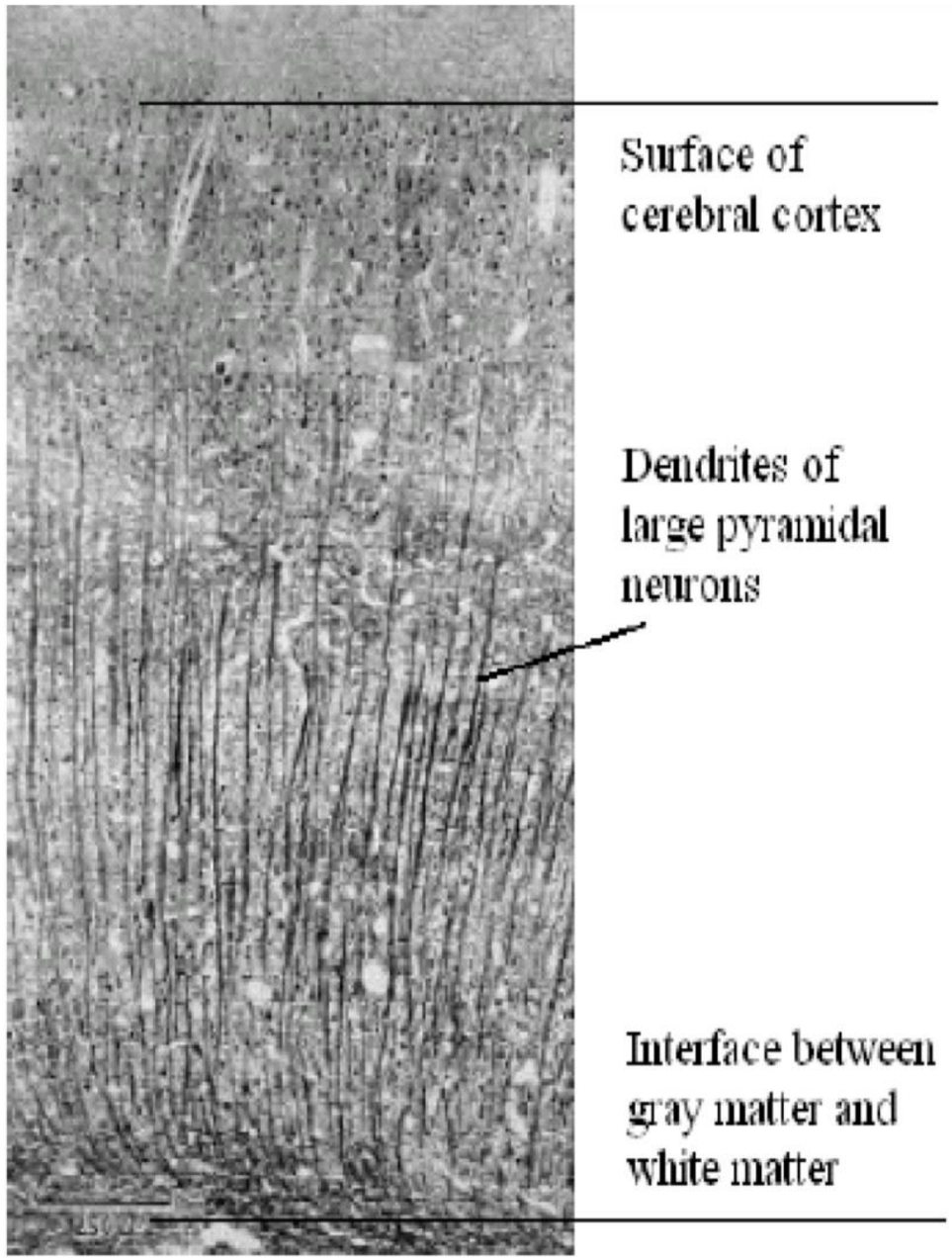
EEG Source

- Dipole from a single neuron is too small
- Neurons in cortex are vertically arranged – can be summated
- EEG is generated by a large number of neurons (cell assembly)

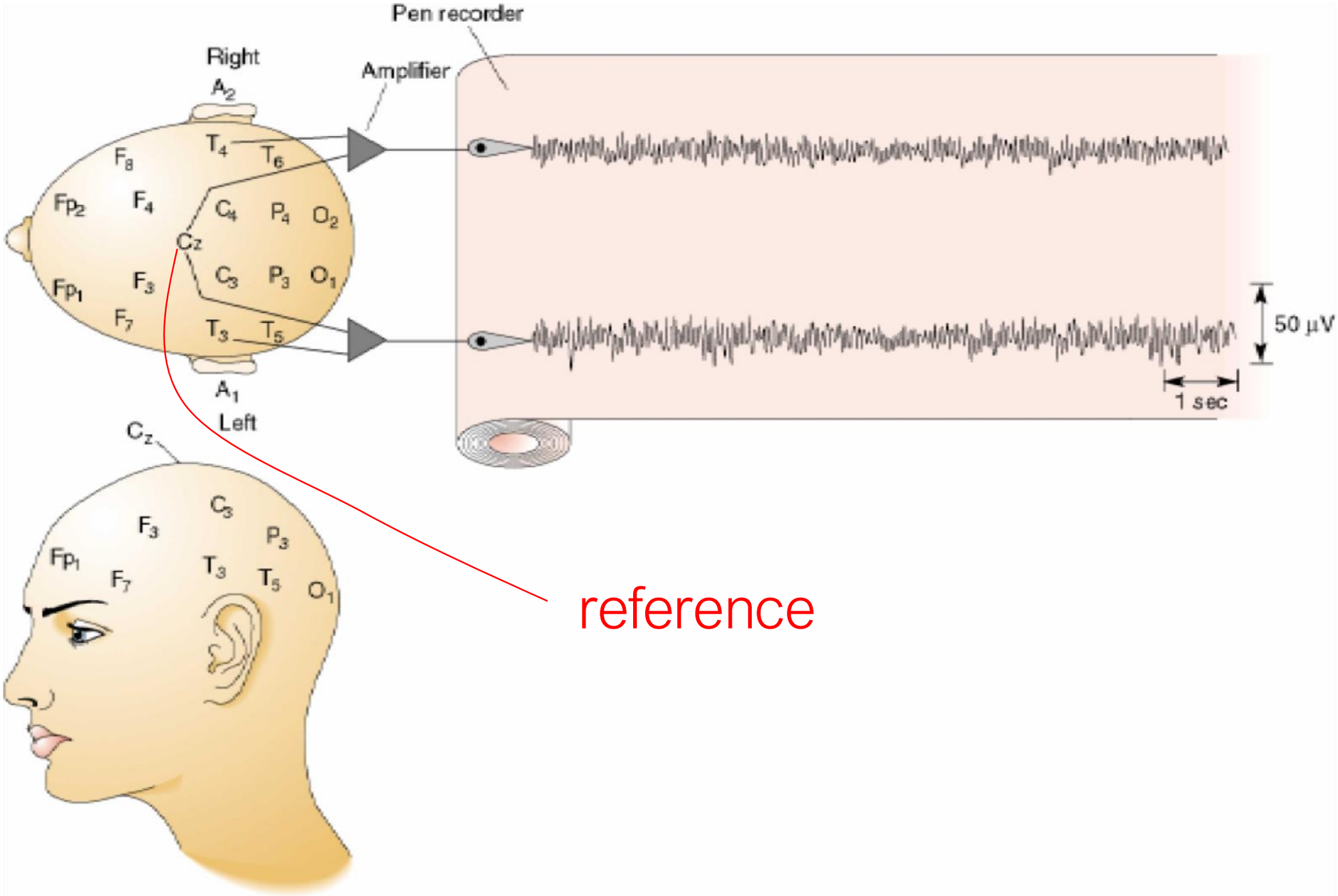


EEG Source

- ~ 6cm^2 cortical surface area is required for scalp EEG recording
- ~10^5 neurons per 0.008cm^2
- So, scalp EEG signal is from ~10^8 neurons

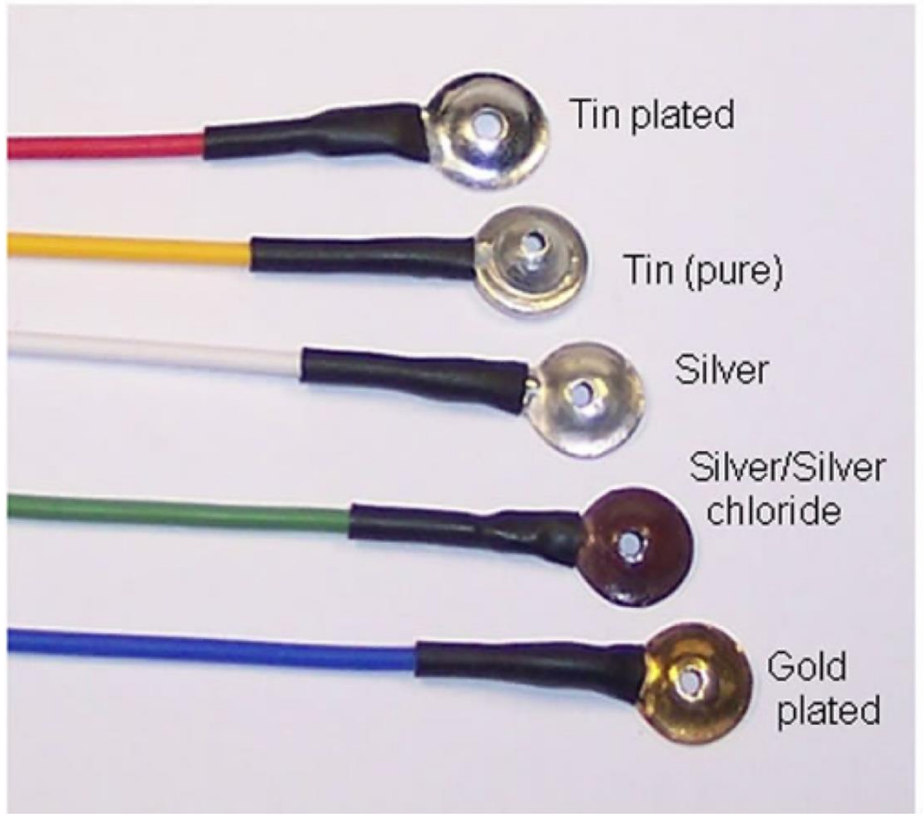


EEG Reference

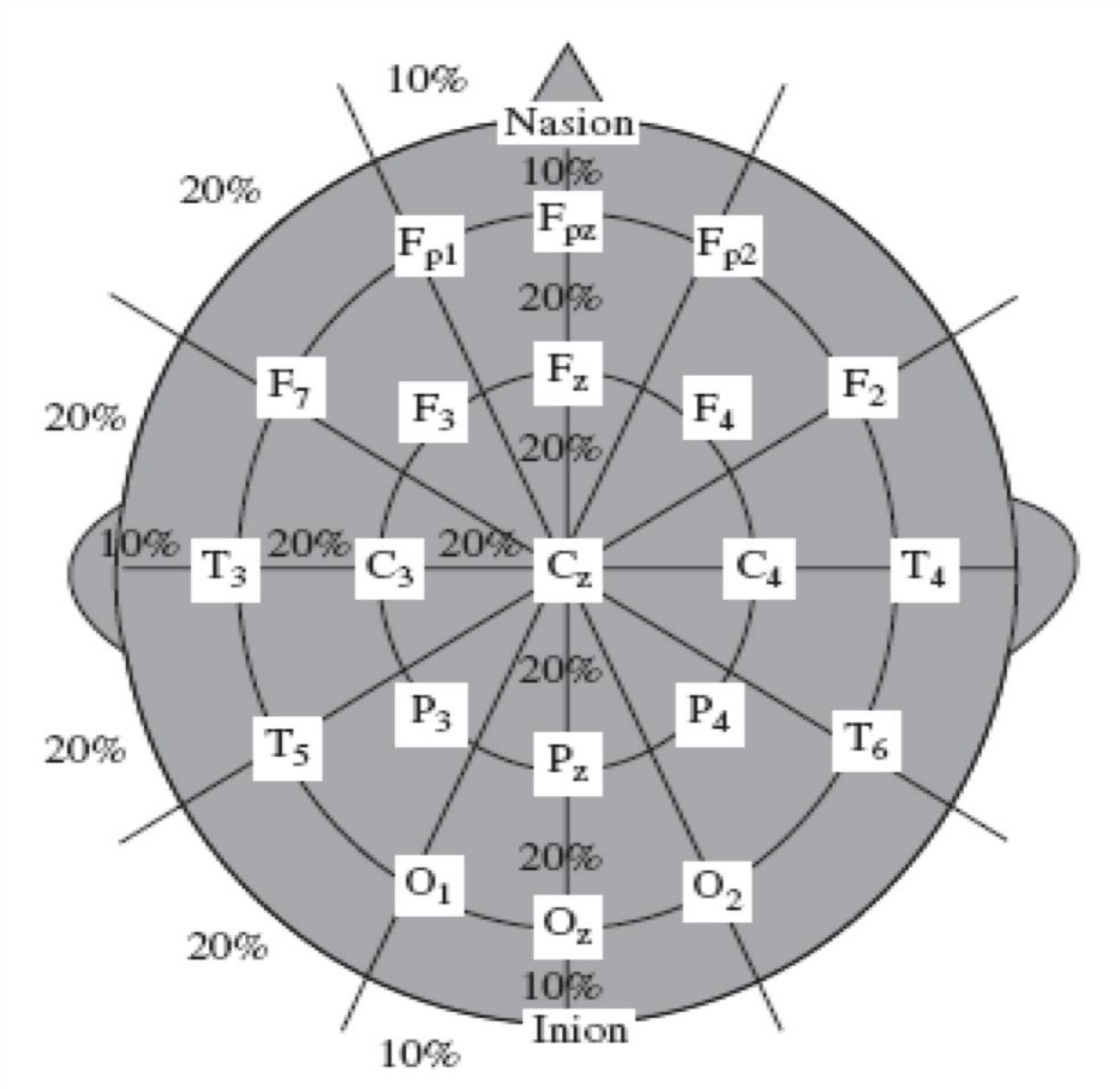


EEG Electrodes

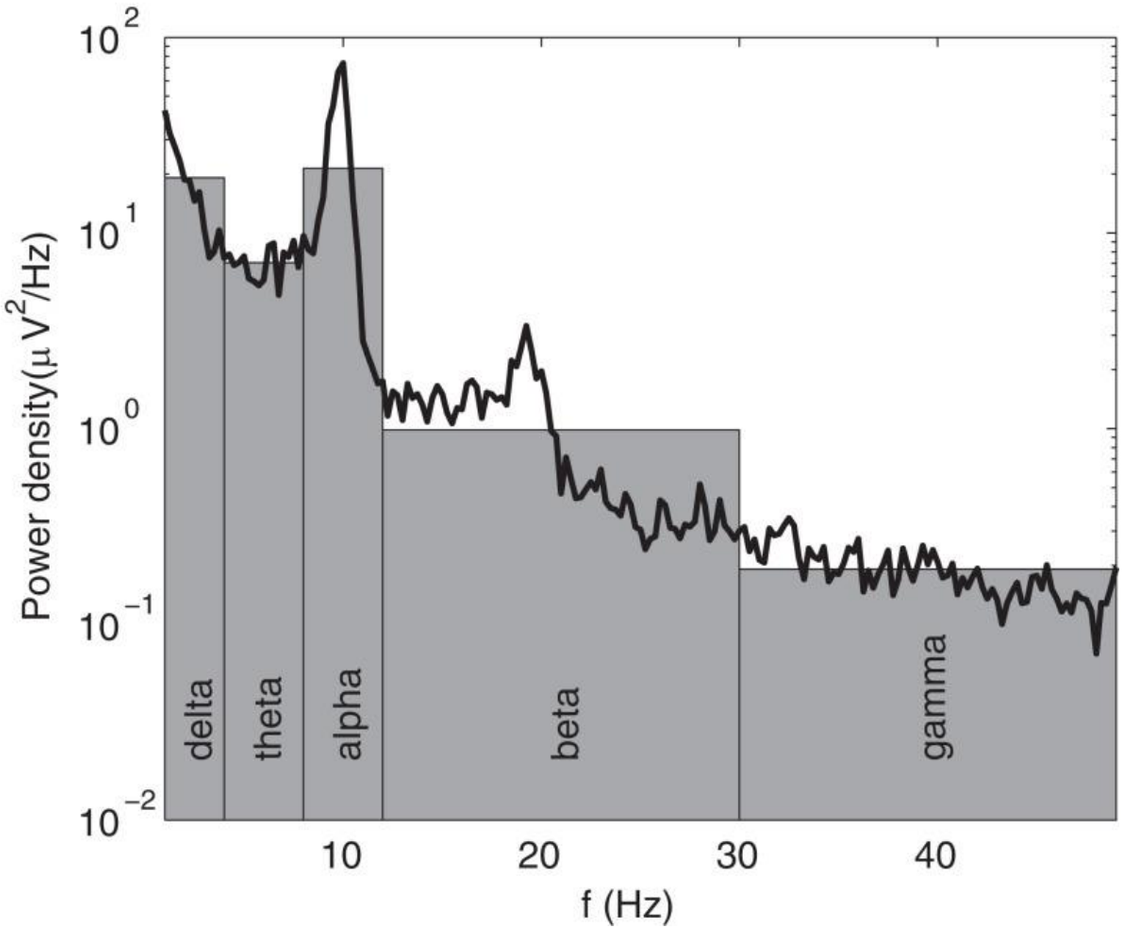
- Analog-to-digital conversion
- Data size: 1-hour recording from 128 channels with 500Hz sampling rate
  - 128 x 3600 x 500 x 16 (bits) ~ 3.68Gbits = 0.45GB
- Common scalp electrodes: Ag-AgCl disks
- Impedance < 5kΩ

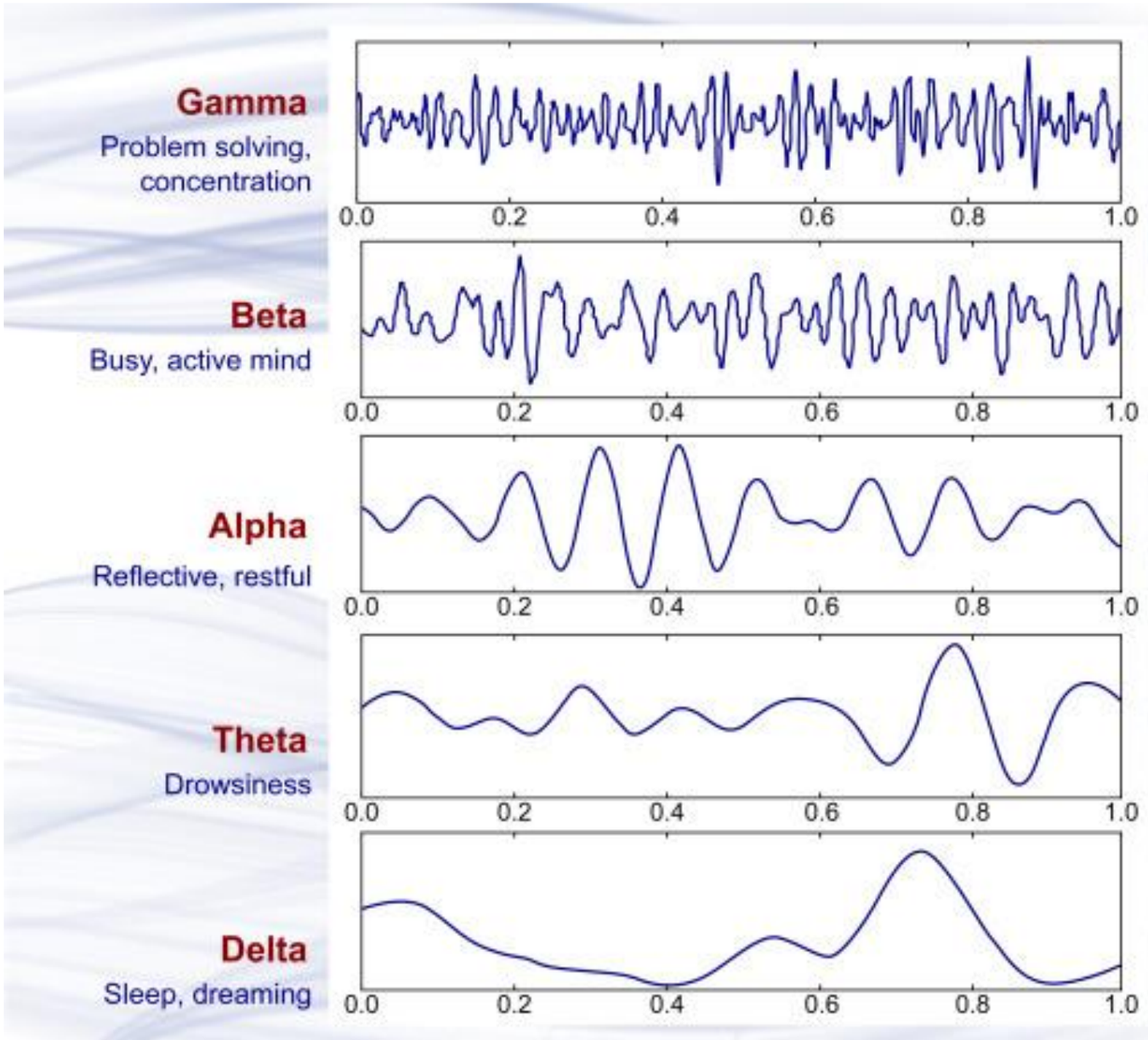


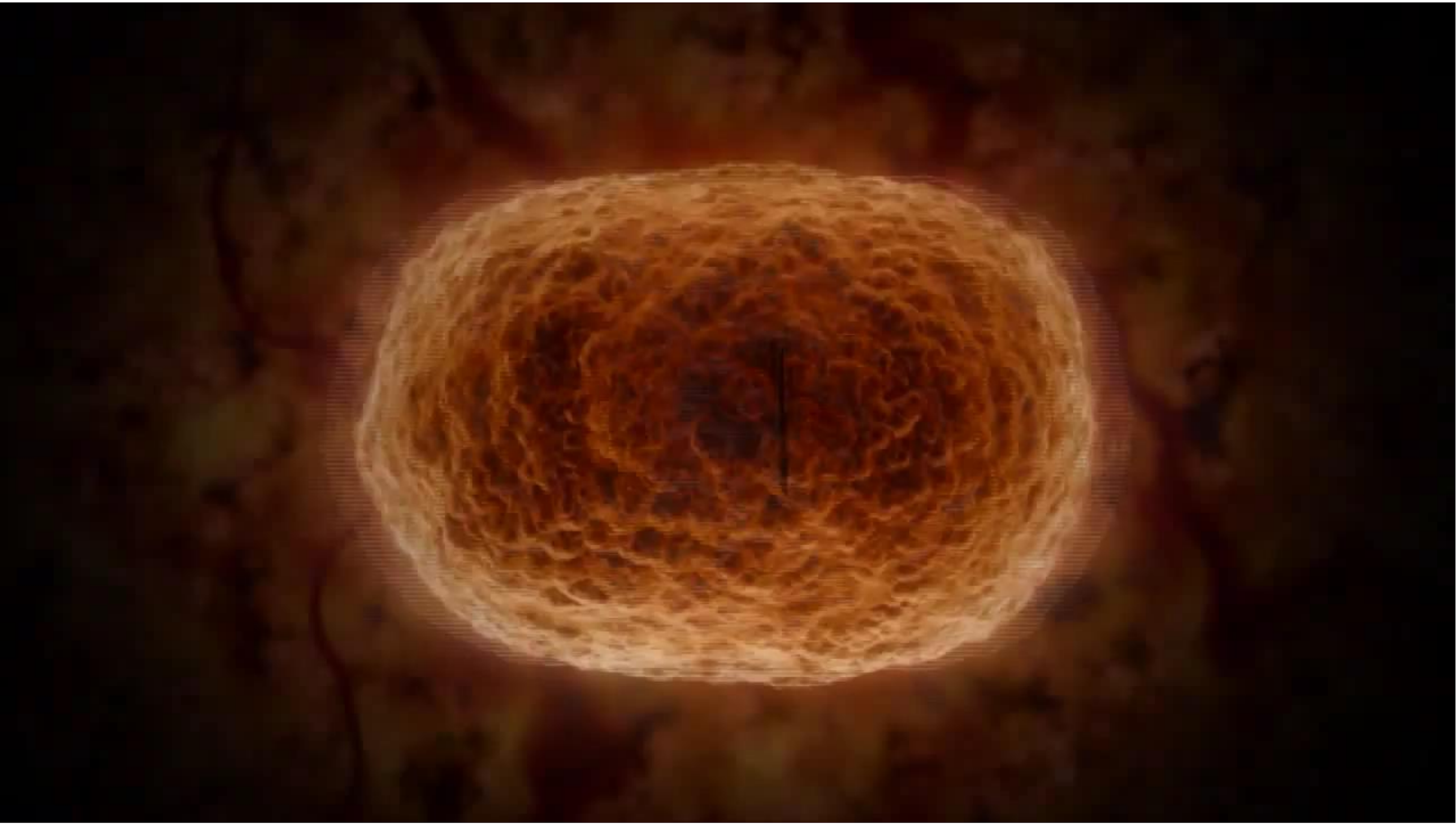
EEG Positioning – 10/20 Standard



EEG Rhythms

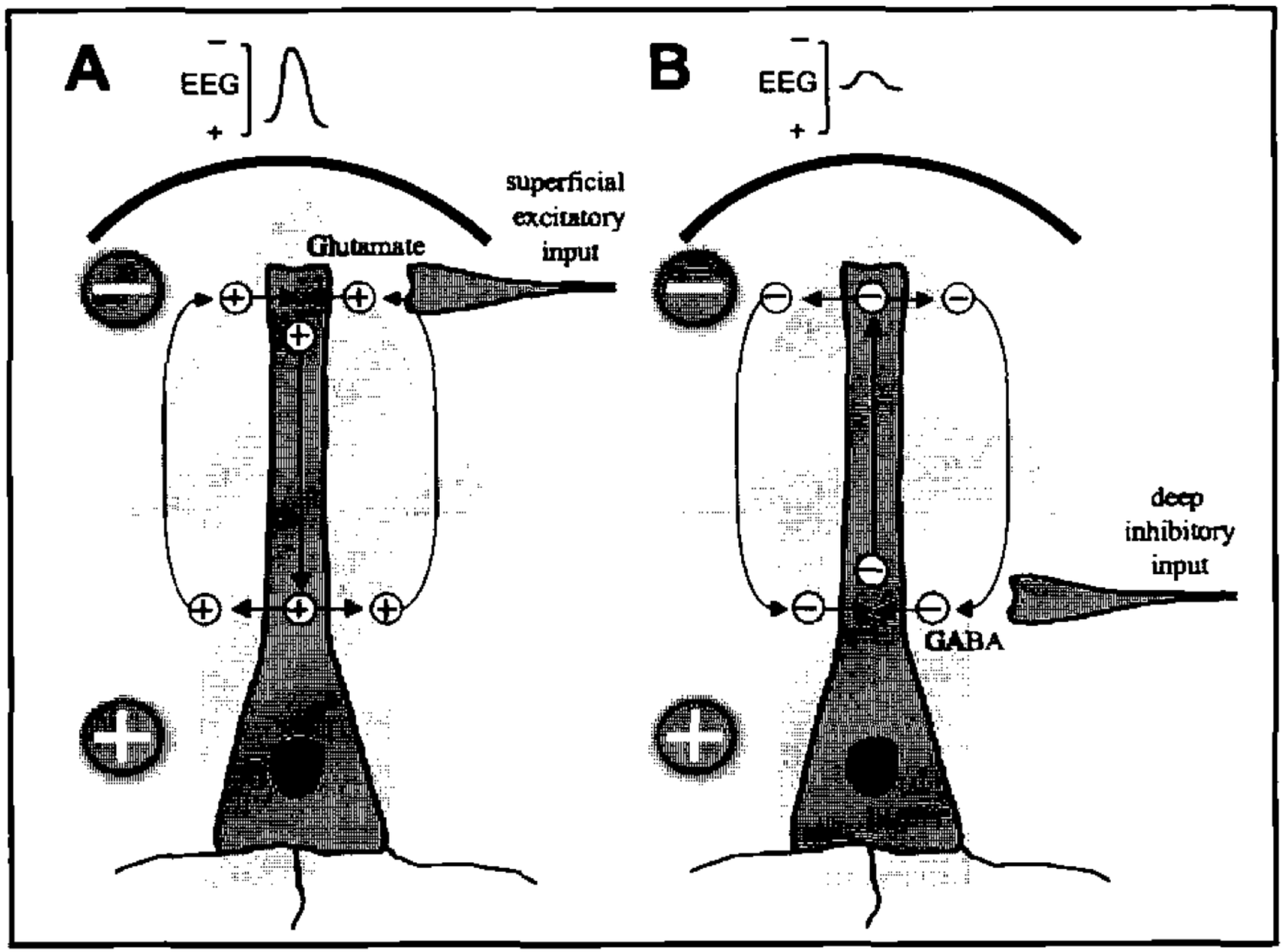

Power density(μ V²/Hz)
10²
10¹
10⁰
10⁻¹
10⁻²
delta
theta
alpha
beta
gamma
10
20
30
40
f (Hz)


Gamma
Problem solving, concentration
Beta
Busy, active mind
Alpha
Reflective, restful
Theta
Drowsiness
Delta
Sleep, dreaming
0.0
0.2
0.4
0.6
0.8
1.0



Source of EEG Deflection

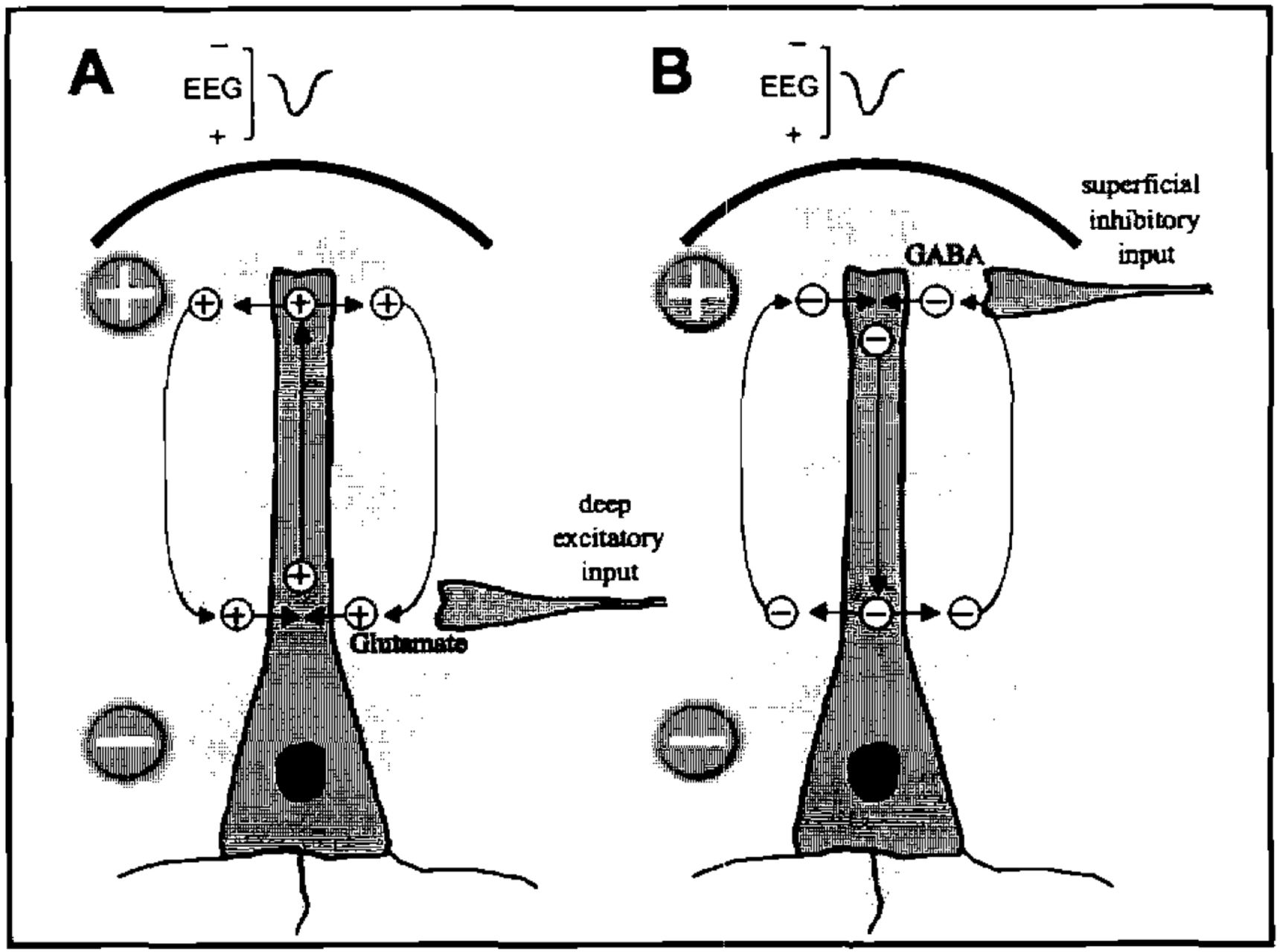
Negative deflection



Kirschstein and Kohling, 2009

Source of EEG Deflection

Positive deflection

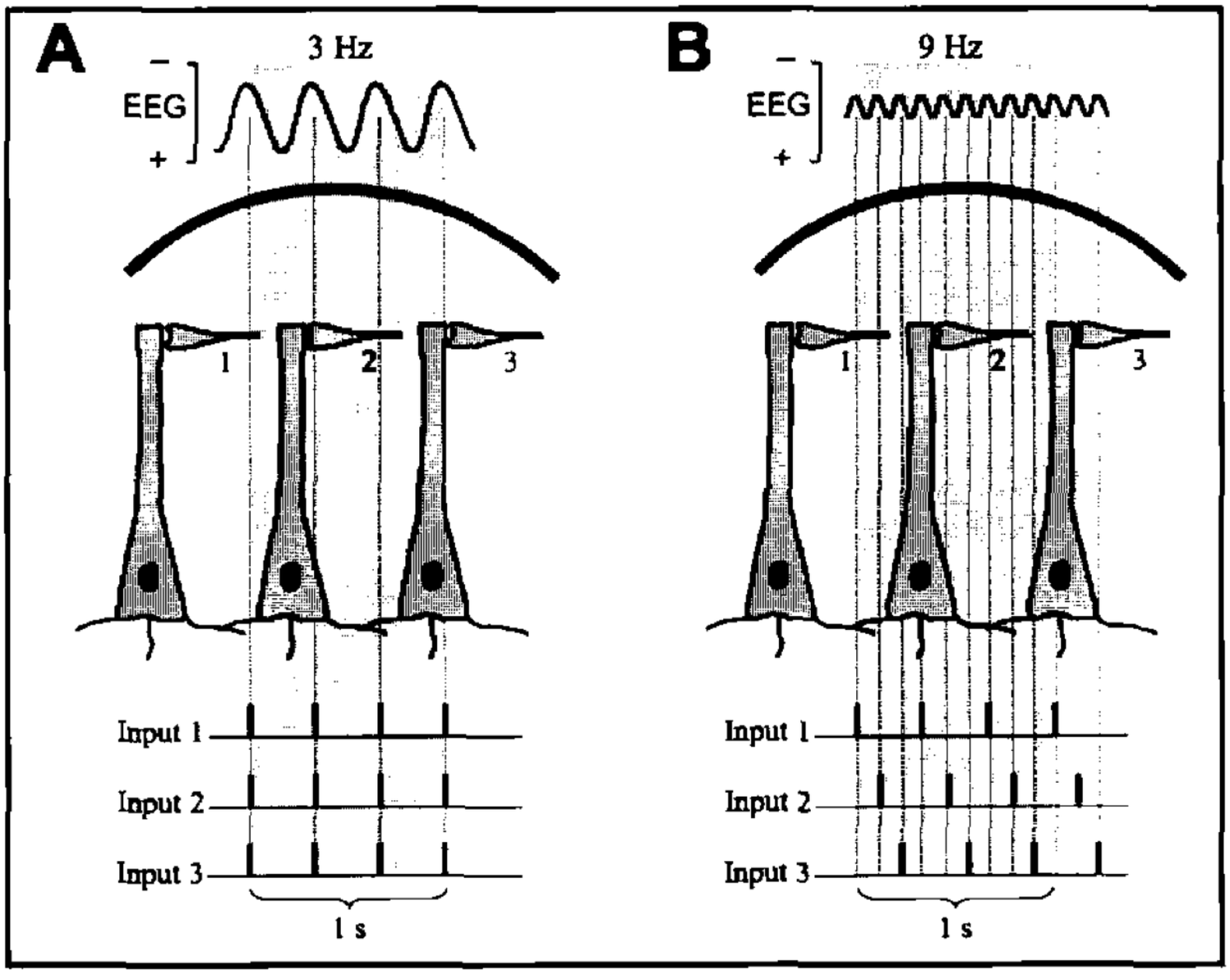


Kirschstein and Kohling, 2009

Source of EEG

Synchronization and Desynchronization

Assumption: superficial excitatory synapses



Kirschstein and Kohling, 2009

EEG Feature Domain

- Time domain
  - Measure: temporal variations of voltage amplitude
  - Example: event-related potentials (ERPs)
- Frequency domain
  - Measure: oscillations at particular frequency
  - Example: mu rhythm, beta rhythm, ...

Outline

- EEG Preprocessing
- Event-Related Potential (ERP)
- Brain Oscillation

- Preprocessing of the EEG signal is an indispensable step for the analysis of EEG in most circumstances

- Is there the standard pipeline of EEG preprocessing? ~ Not yet

- Preprocessing ~ any necessary digital signal processing operations to polish up raw EEG signals in order to leave only brain activity signals for subsequent analyses

- Preprocessing also involves procedures to **enhance** spatiotemporal characteristics of the EEG signal related to the task

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wiki tools

Makoto's preprocessing pipeline

If you have said 'EEG data mining is like walking a mine field,' you are a friend of mine. It's a field of mine because it's not a field of mine. But never mind even if you have never mined EEG data. I'll show you my minecraft. This is Makoto Miyakoshi's personal recommendation for preprocessing EEG data using EEGLAB. These are NOT Swartz Center's official opinions. This page is admittedly dogmatic, so follow the instruction at your own risk. This is a forever beta version and subject to change anytime. If you have any question, please post it to EEGLAB mailing list (you need to subscribe it first.)

Note from Arnaud Delorme (main developer of EEGLAB): See also this page for a quick pipeline to reject ICA components in single datasets.

Contents [hide]

1 Change the option to use double precision
2 Import data
3 Downsample if necessary
4 High-pass filter the data at 1-Hz (for ICA, ASR, and CleanLine)(03/29/2017 updated)
5 Import channel info
6 Remove bad channels
7 Interpolate all the removed channels
8 Re-reference the data to average (11/29/2017 updated)
9 Remove line noise using CleanLine (04/13/2018 Updated)
10 Epoch data to -1 to 2 sec (12/07/2016 updated)
11 Reject epochs for cleaning (03/29/2018 Updated)
12 Adjust data rank for ICA (12/26/2016 updated)
13 Run ICA (07/25/2017 updated)
    13.1 To learn how to evaluate EEG and artifact ICs (01/24/2017 updated)
    13.2 Unofficial instruction to install AMICA (08/03/2017 updated)
14 Estimate single equivalent current dipoles
15 Search for and estimate symmetrically constrained bilateral dipoles
16 Create STUDY (01/05/2017 updated)
    16.1 A tip to compute time-frequency transform (i.e. ERSP & ITC) (01/11/2017 updated)
17 Alternatively, cleaning continuous data using ASR (03/16/2018 updated)
    17.1 What is Artifact Subspace Reconstruction (ASR)?
    17.2 Download

EEG-Clean-Tools (PREP Pipeline)

Contains tools for EEG standardized preprocessing

View on GitHub | Download .zip | Download .tar.gz

Introduction to the PREP pipeline

The PREP pipeline is a standardized early-stage EEG processing pipeline that focuses on the identification of bad channels and the calculation of a robust average reference. PREP also has an extensive reporting facility. It is designed to be run in a completely automated way. The major sections of this document are: * Introduction (requirements, citing, installation) * Algorithm (steps, meaning of parameters for each step) * Running as an EEGLAB plug-in * Running as a script

Why preprocessing?

- Preprocessing influences the subsequent data analysis results

- Example:

  - Classification of different mental states from EEG via a BCI depends on how EEG preprocessing treated the recorded EEG signals

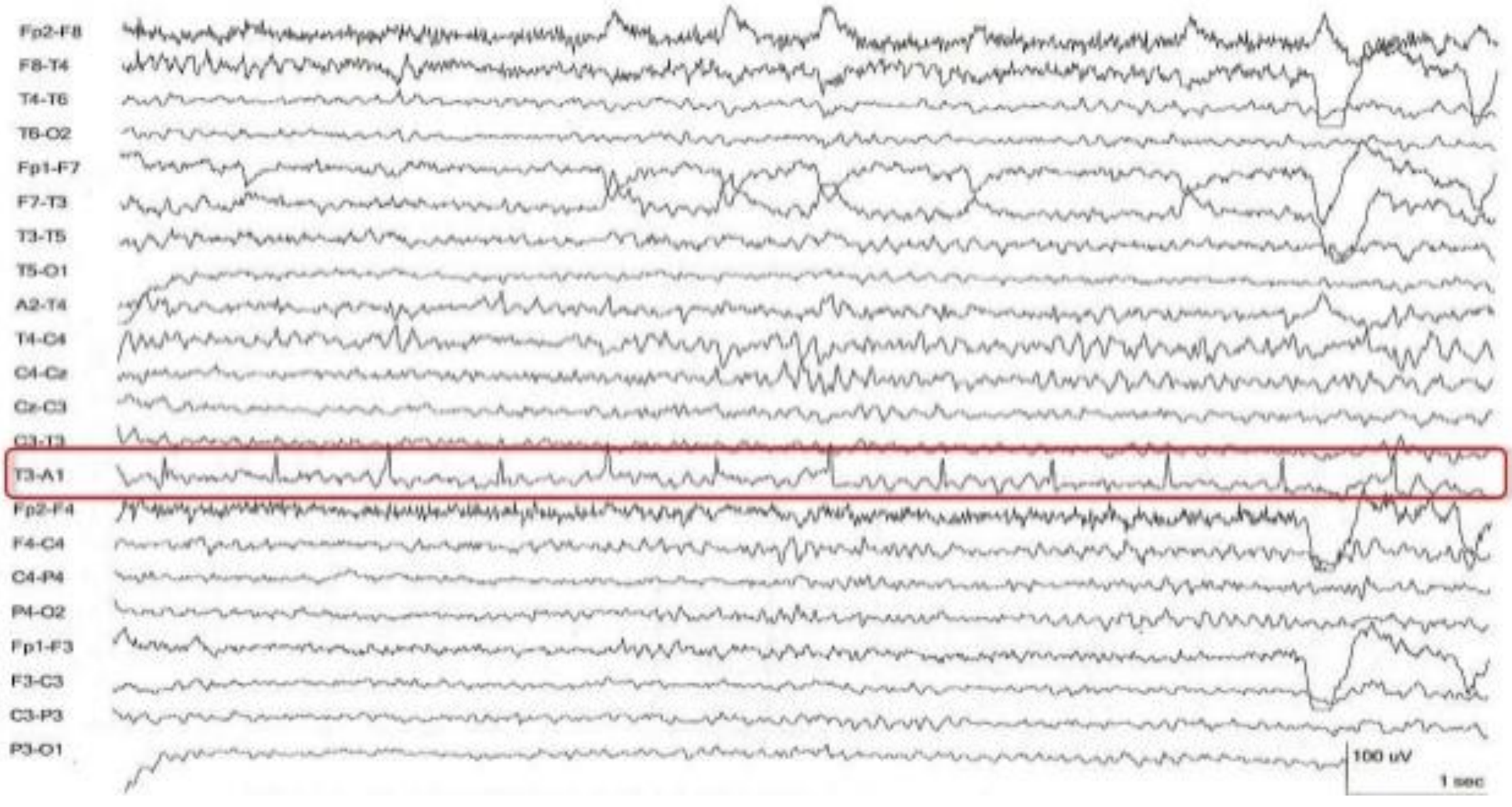
- Any analysis result from EEG data containing significant noise and artifacts would draw misleading conclusions

- Need the standardization of preprocessing routines for multi-site data collection in divergent experimental environments

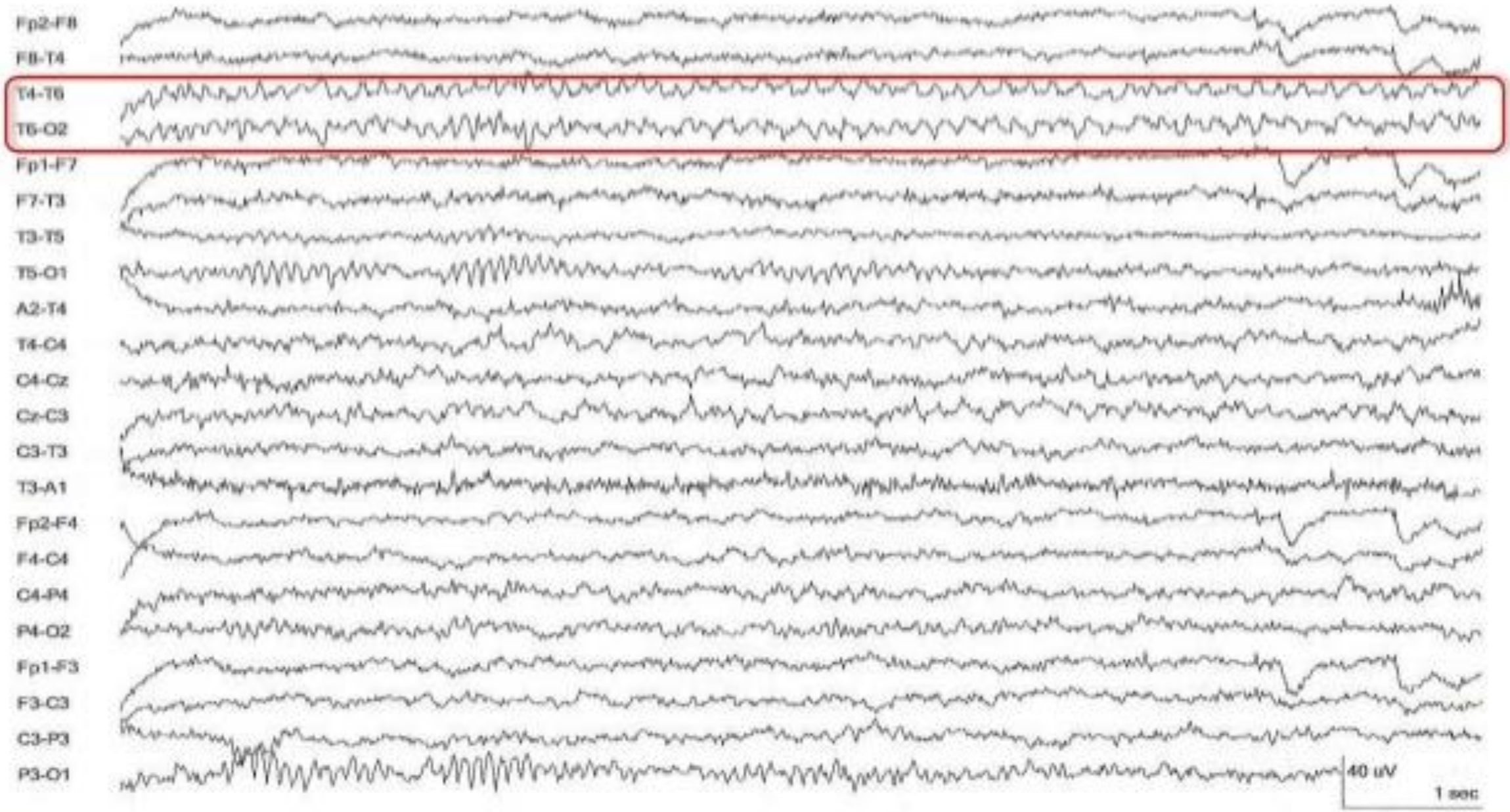
- Preprocessing involves the removal of any unnecessary covert and overt components of EEG
  - Unnecessary components
    - Noise: neurological activities irrelevant to an examined behavioral task
    - Artefact: from external sources unrelated to neurological activities, such as eye movements, respiration or electrical interference
- In this class, we will focus on artefact removal

Examples of Artefacts

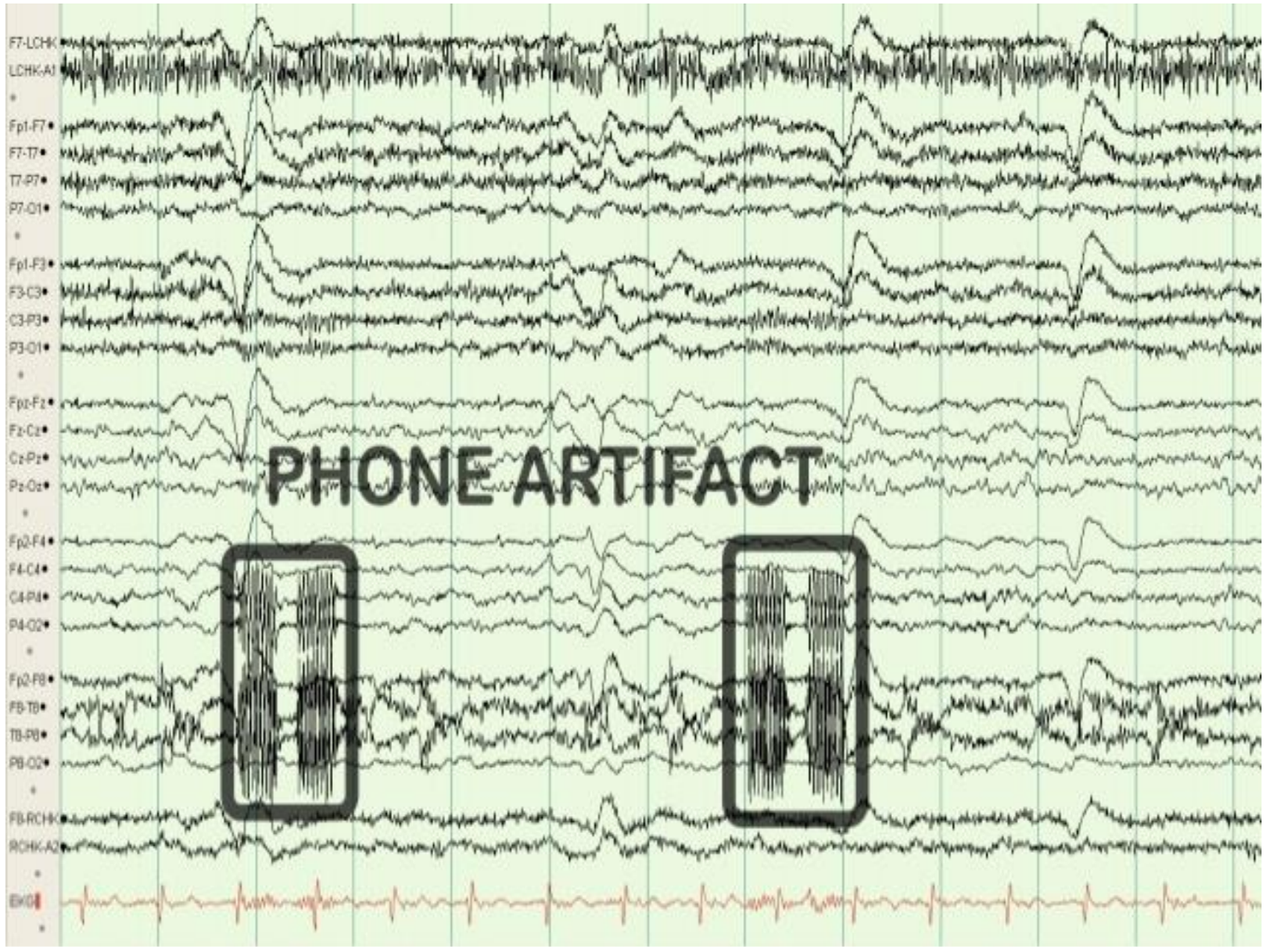
Cardiac Artefact



Electrode Movement Artefact



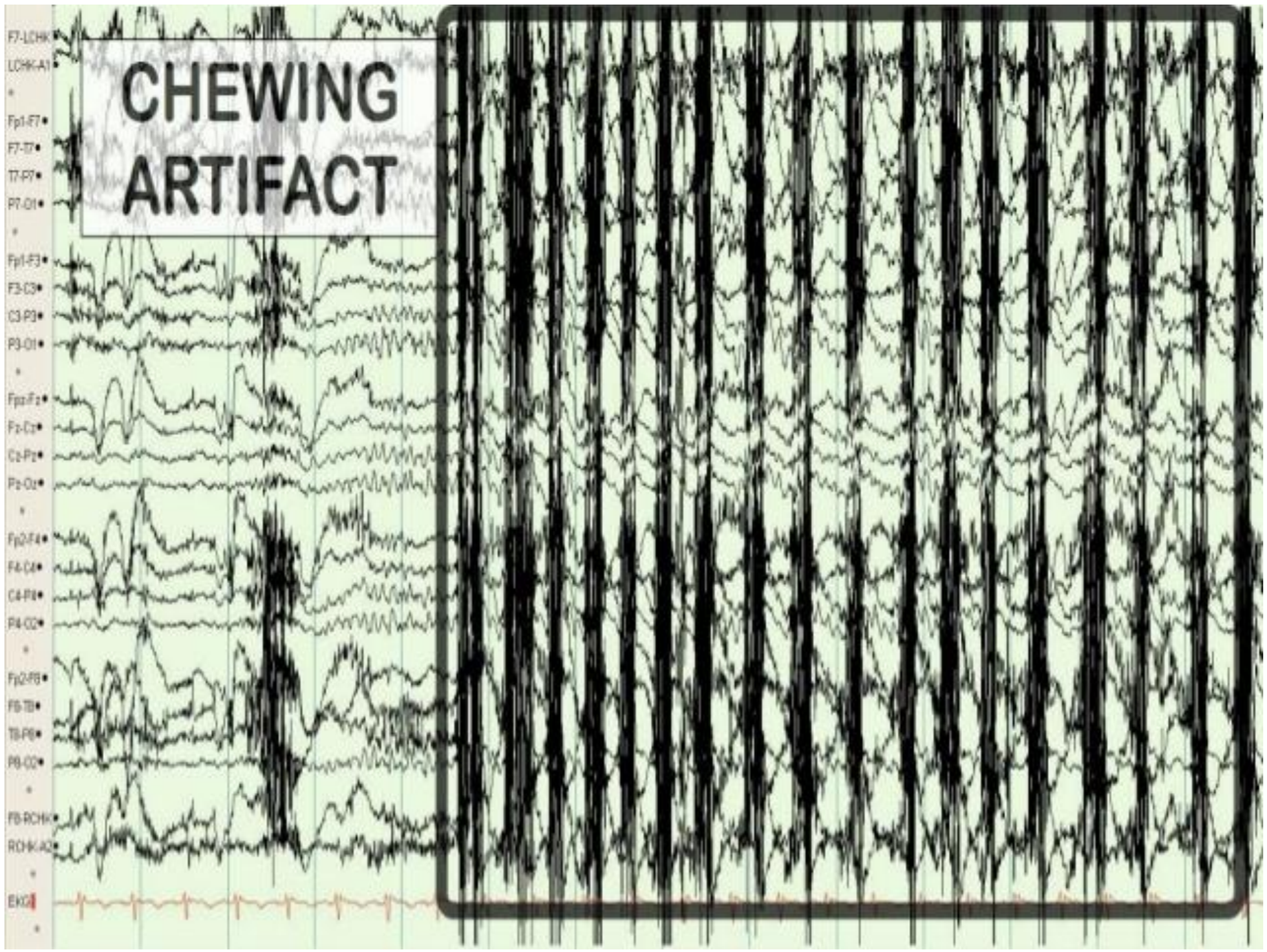
Phone Artefact



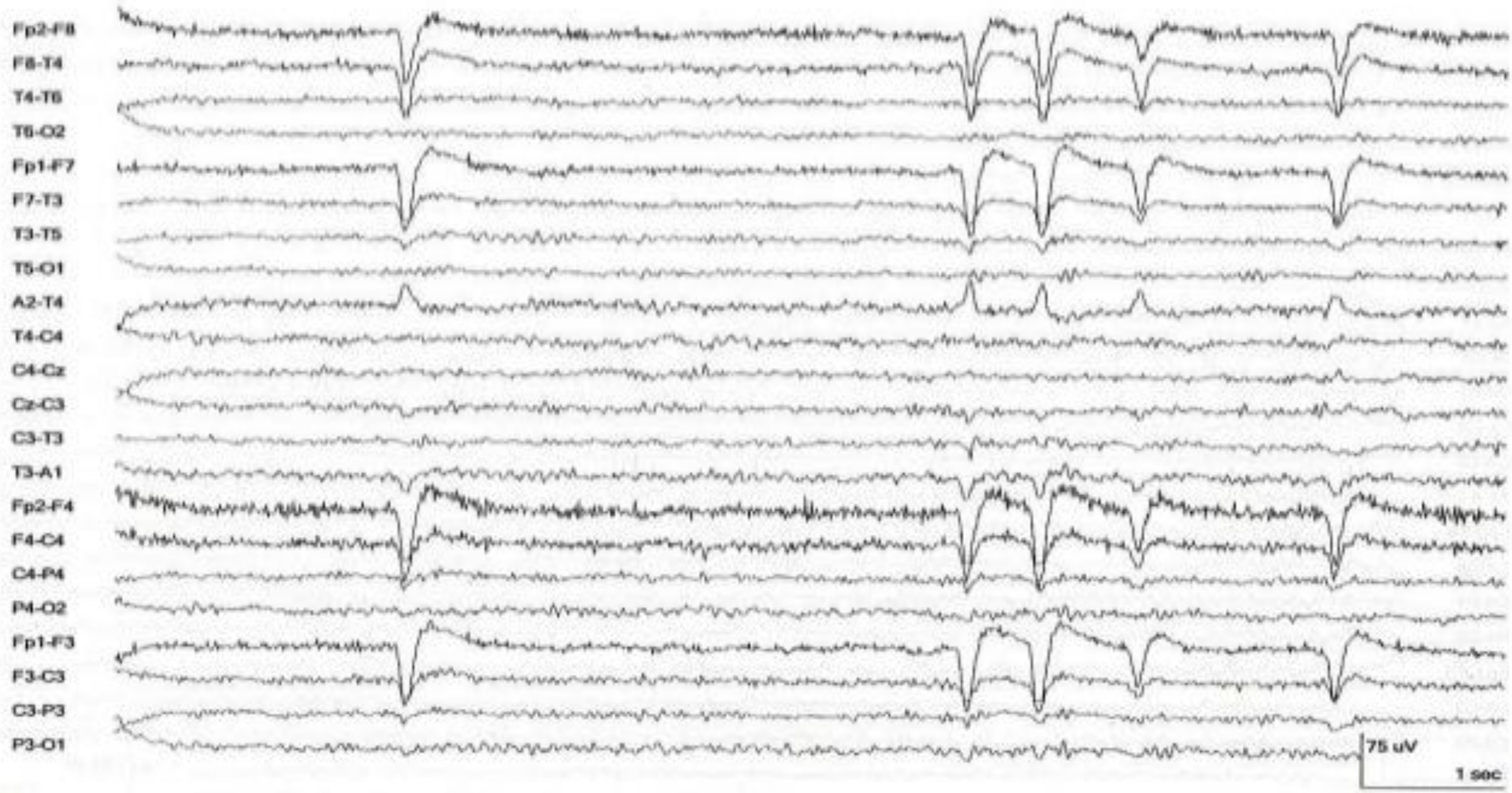
Facial Muscle Artefact



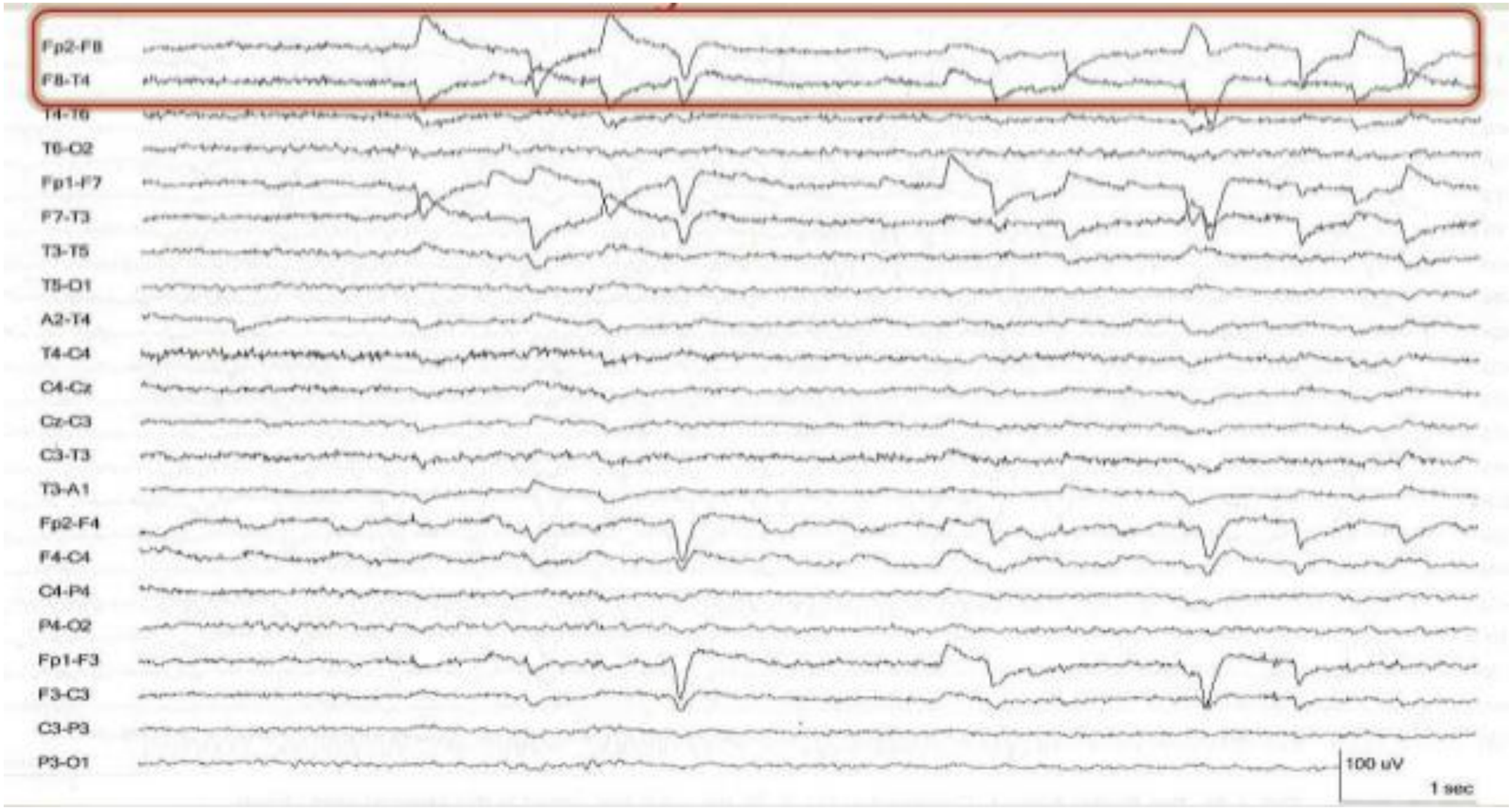
Chewing Artefact



Blink Artefact



Lateral Eye Movement Artefact



Early-State Preprocessing

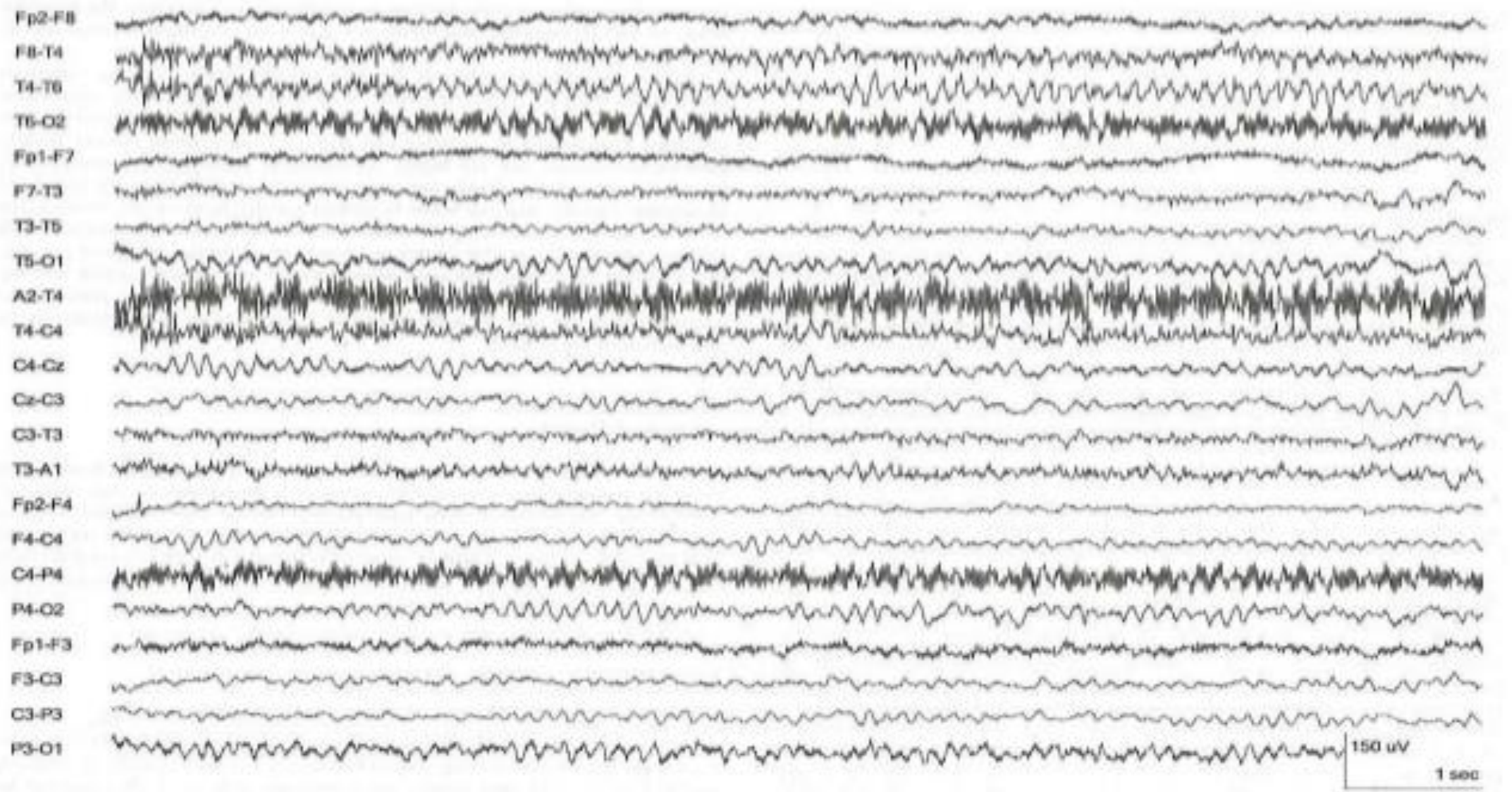
Early-Stage Preprocessing

Line Noise Removal

Re-referencing

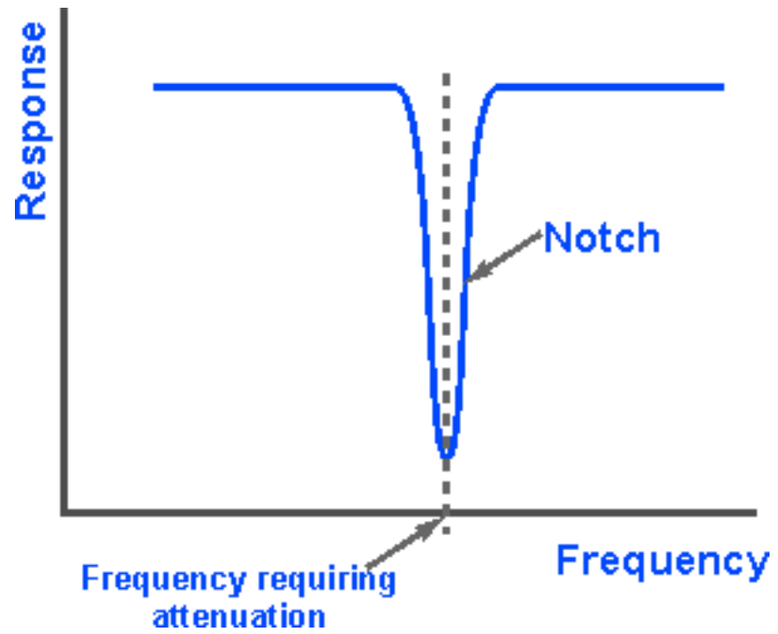
Bad Channel Elimination

Line Noise

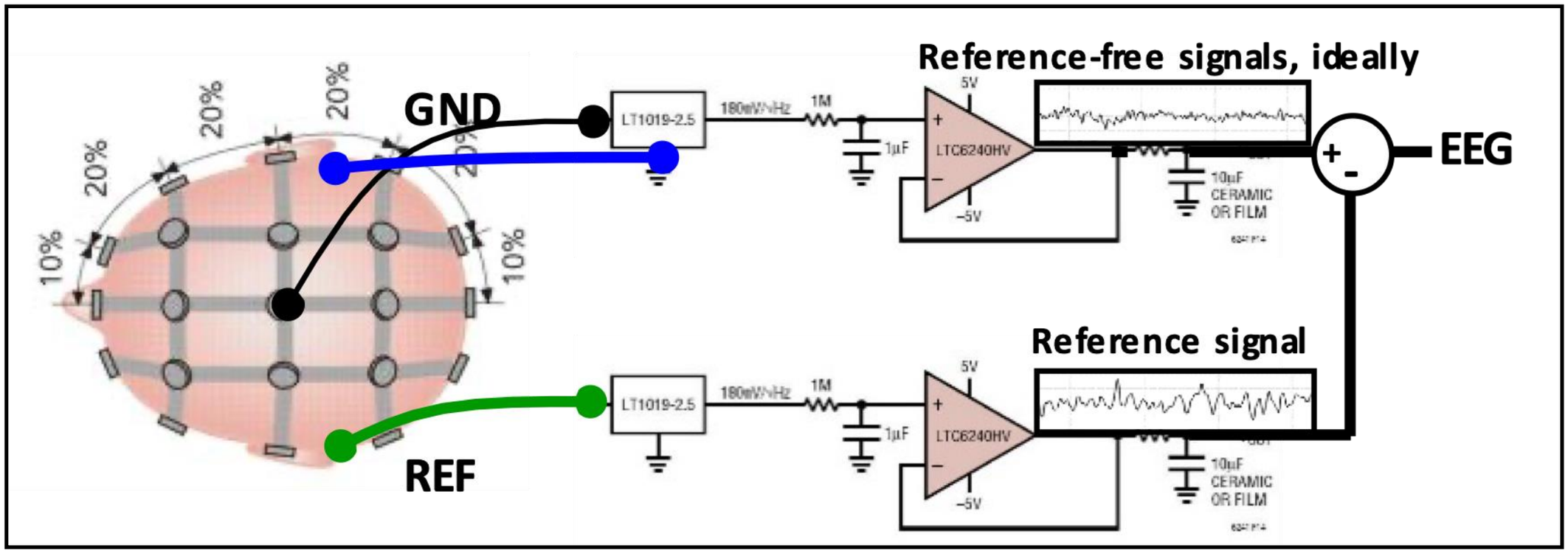

Fp2-F8
F8-T4
T4-T6
T5-O2
Fp1-F7
F7-T3
T3-T5
T5-O1
A2-T4
T4-C4
C4-Cz
Cz-C3
C3-T3
T3-A1
Fp2-F4
F4-C4
C4-P4
P4-O2
Fp1-F3
F3-C3
C3-P3
P3-O1
150 uV
1 sec

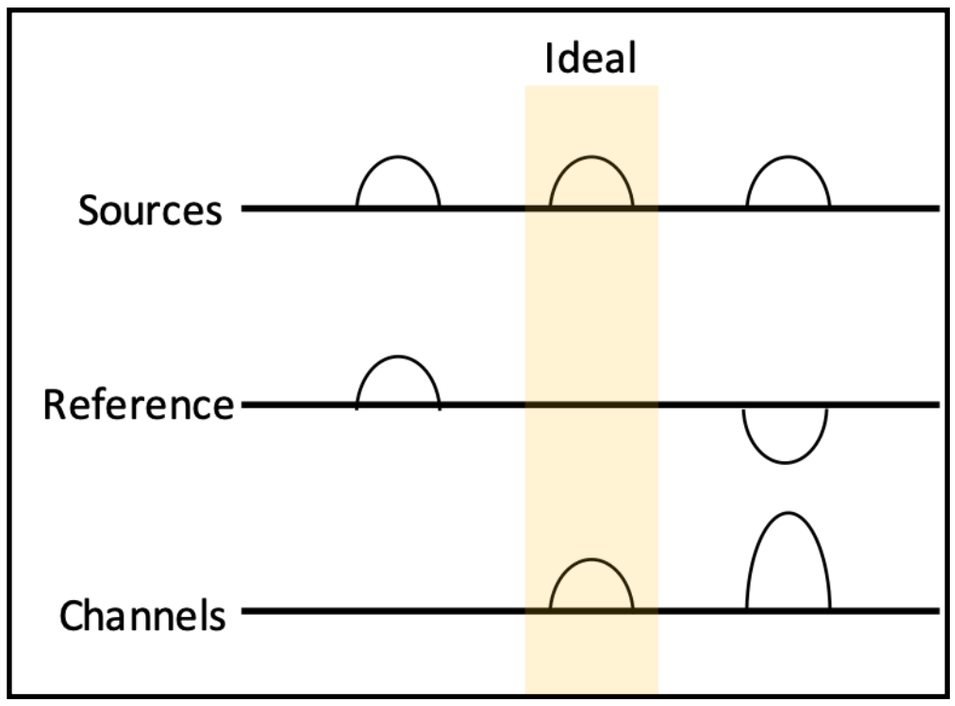
Line Noise Removal

- Most popular method: notch filtering
  - Implement a notch filter at 60 ± 10 Hz
- Problems
  - Unintended distortions in oscillations between 50 ~ 70 Hz
- Remedy
  - Follow-up low-pass filtering < 50 Hz → Can alter temporal structure of EEG

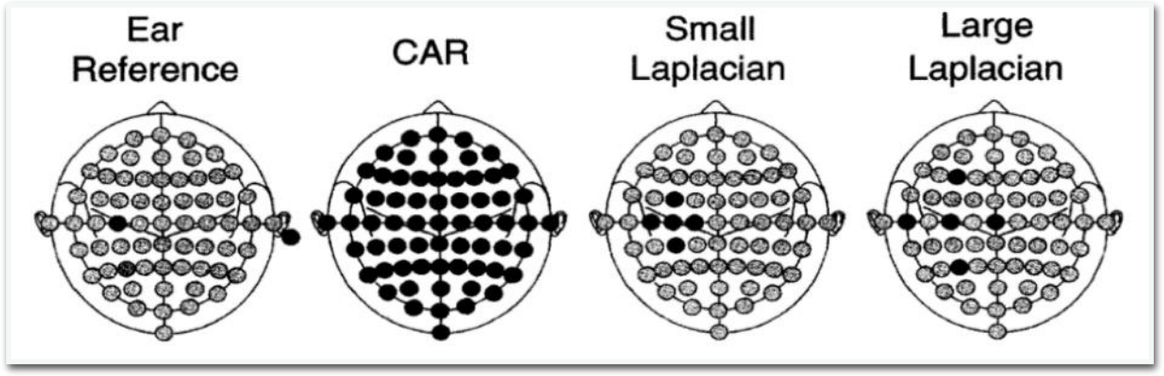


Re-referencing





Conventional Re-Reference Methods



McFarland et al., 1997

Bad Channel Elimination

- “Bad” channel ~ a channel having contaminated EEG signals
- Screen each channel to find EEG signals with excessively large amplitudes

Bad Channel Elimination

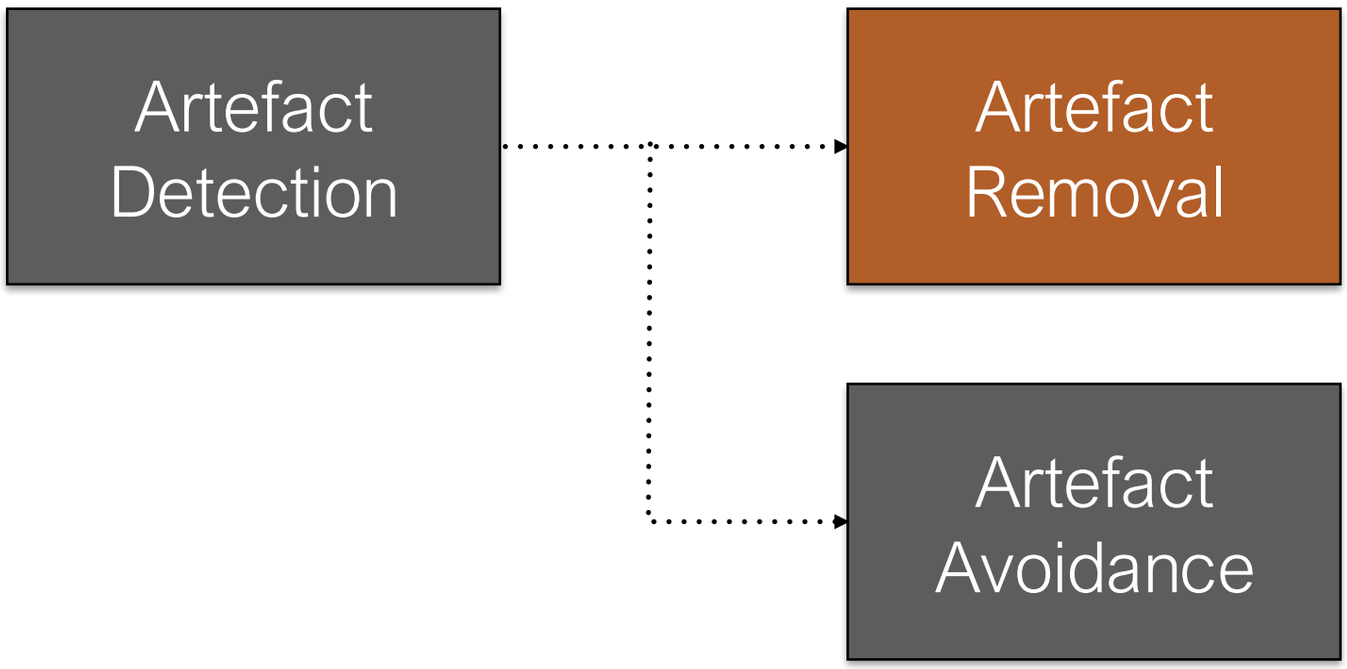
- Detection of a bad channel
  - the standard deviation of z-transformed signal > a threshold
  - correlation of a single channel with others
    - Normal EEG shows across-channel correlations in the low-frequency
    - Abnormal EEG: across-channel correlation after lowpass filtering < threshold (e.g. 0.4)

Bad Channel Elimination

- Interpolation
  - Replace bad channels with virtual healthy channels: created by the interpolation from neighboring channels
  - Interpolation methods:
    - spherical splines, higher-order polynomials, nearest-neighbor averaging, radial basis function, …

Late-State Preprocessing

Artifact Removal



Artefact Removal Methods

- To cancel artefacts from EEG with minimal distortions in the brain signal
- Single-channel method
  - Process each channel independently
    - Linear regression, adaptive filtering, wavelet transform, …
- Multi-channel method
  - Process whole channels together
    - Blind source separation, …

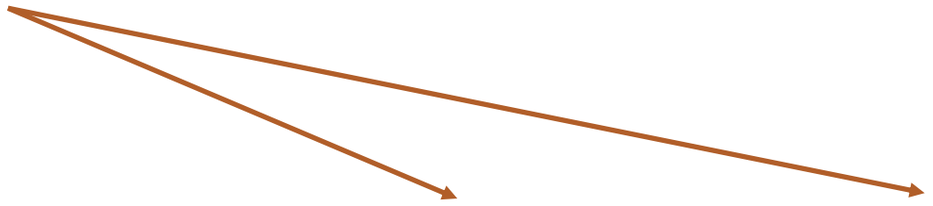
Linear Regression

- Simple and easy to use

- Estimate a portion of EEG contaminated by artefacts using regression → Subtract the regressed portion from the contaminated EEG

- Assumption: EEG = Original brain signal + a fraction of the artefact represented in reference

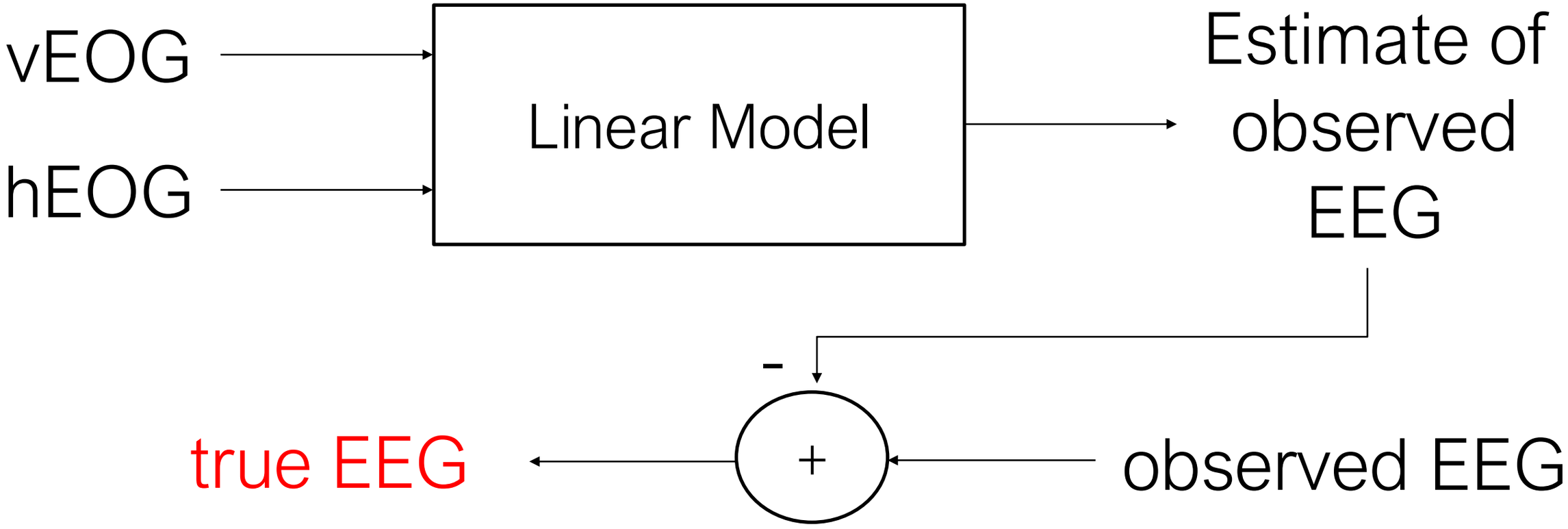
- Estimate the fractional factor from both the observed EEG signal and reference channel



observed EEG = true EEG + b_1vEOG + b_2hEOG

Linear Regression

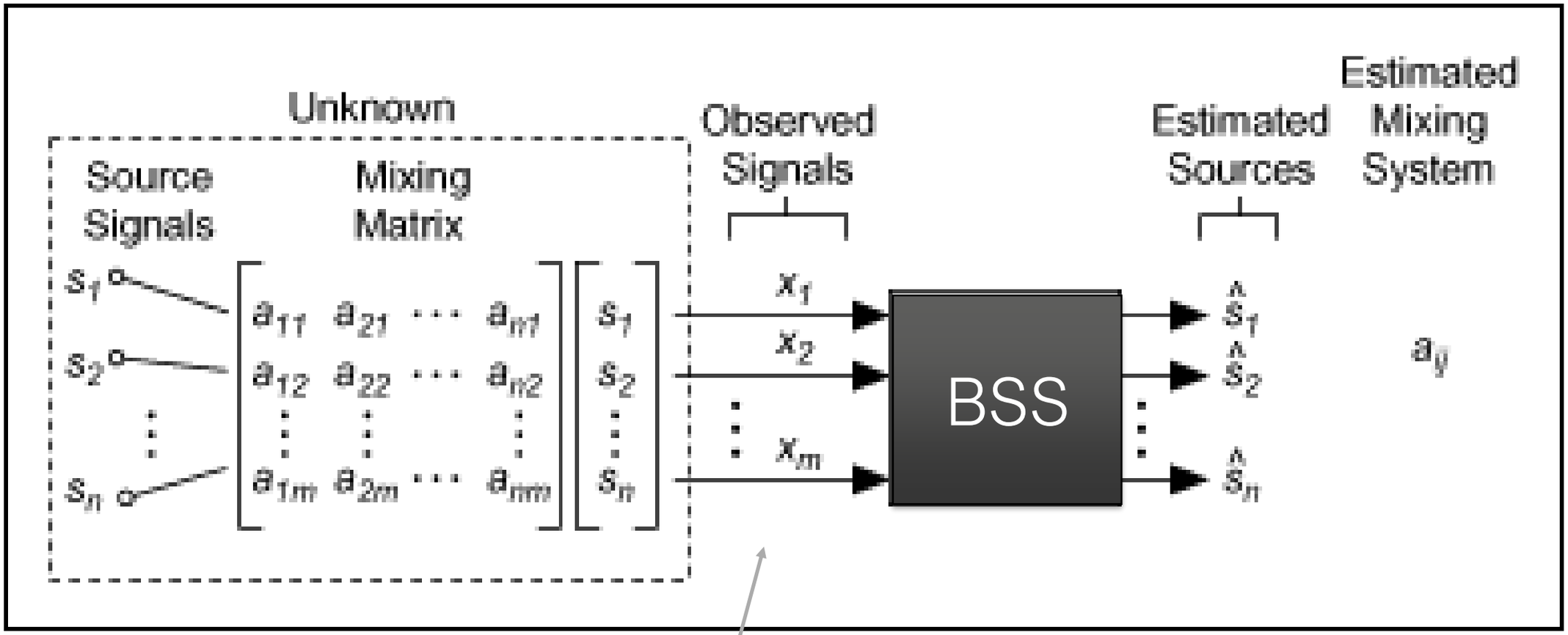
observed EEG = true EEG + b_1vEOG + b_2hEOG



Linear Regression

- Drawbacks
  - One or more reference channels must be available
    - Or can be estimated from EEG
  - Linearity assumption: a linear combination of EEG and artifacts
  - Only applies well to a few types of artifacts (e.g. EOG, ECG)
  - Bidirectional contamination
    - subtraction of a regressed portion of ocular artifacts from EEG can also take out cerebral components

Blind Source Separation

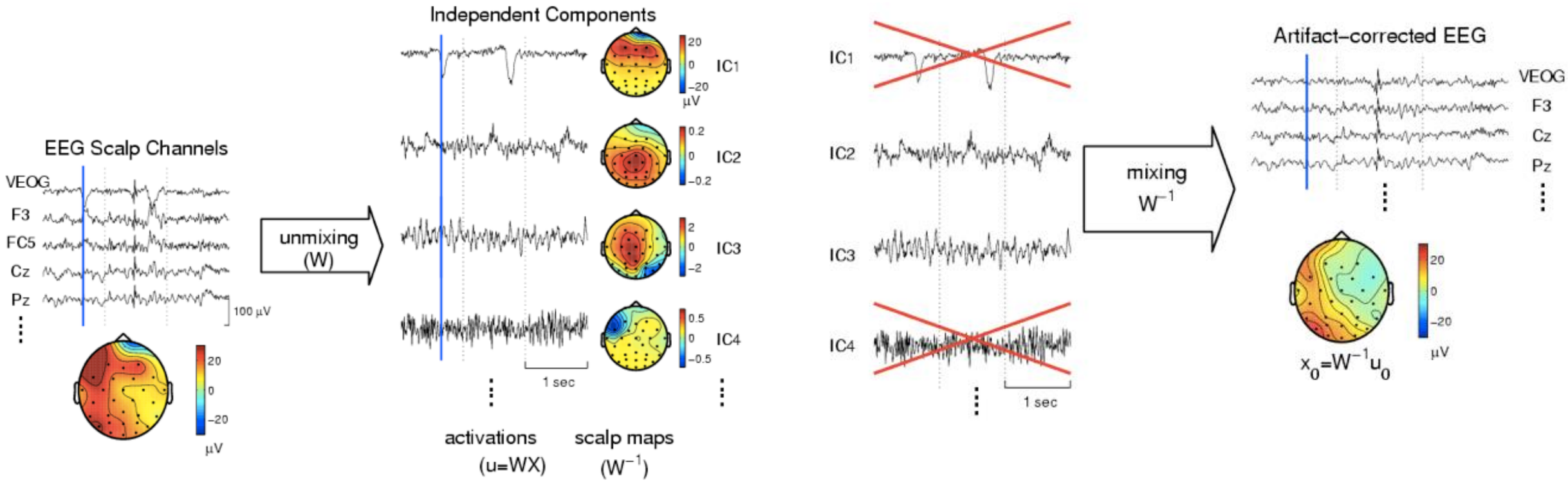


$$x = As \qquad \hat{s} = Wx$$

$$W = A^{-1}$$

(National Instrument)

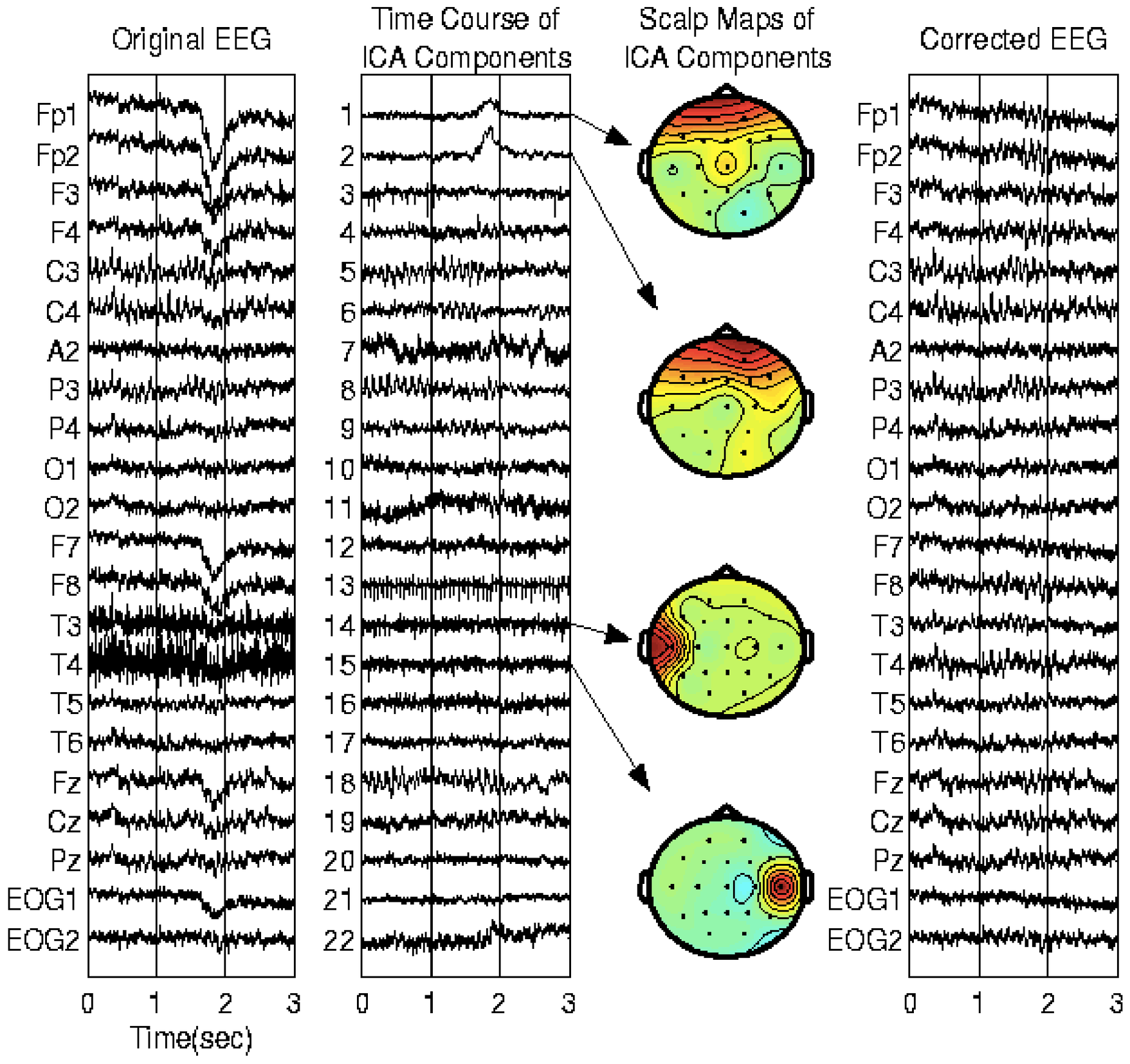
Blind Source Separation



(Jung and Makeig)

Blind Source Separation

- BSS methods applied to EEG artefact removal
  - Independent Component Analysis (ICA)
  - Principal Component Analysis (PCA)
  - Canonical Correlation Analysis (CCA)
  - Morphological Component Analysis (MCA)
  - Singular Spectrum Analysis (SSA)



(Jung and Makeig)